

信号处理原理-04

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回顾——信号的正交分解

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➤ 信号（函数）的正交分解

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\langle \varphi_i(t), \varphi_j(t) \rangle = 0 \quad \Rightarrow \quad f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$\varphi_j(t)$

$$c_j = \frac{\langle f(t), \varphi_j(t) \rangle}{\langle \varphi_j(t), \varphi_j(t) \rangle} = \frac{1}{k_j} \langle f(t), \varphi_j(t) \rangle$$

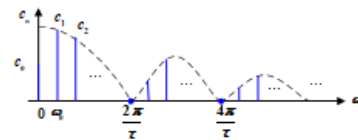
$$\langle f(t), \varphi_j(t) \rangle = \sum_{i=1}^{\infty} c_i \langle \varphi_i(t), \varphi_j(t) \rangle \quad \Rightarrow \quad \langle f(t), \varphi_j(t) \rangle = c_j \langle \varphi_j(t), \varphi_j(t) \rangle$$

回顾——周期信号的傅立叶级数

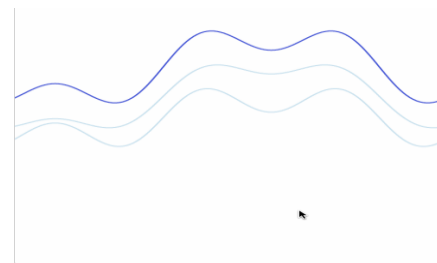
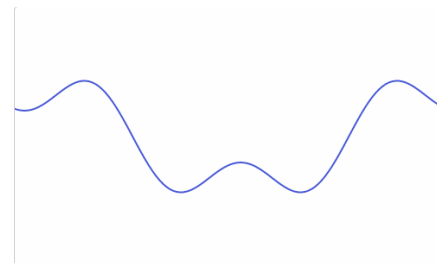
2

➤ 傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



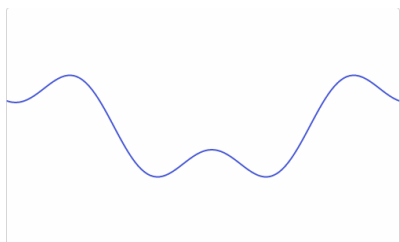
$$\begin{cases} a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt \\ a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt \\ b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt \end{cases}$$



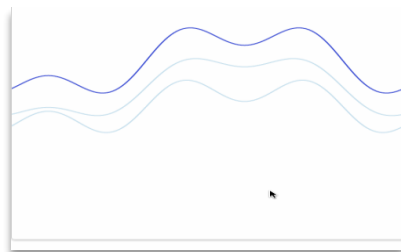
回顾——周期信号的傅立叶级数

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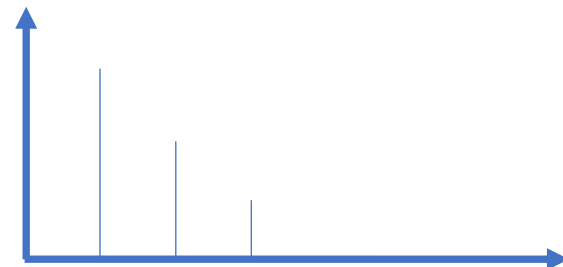
➤ 用途？



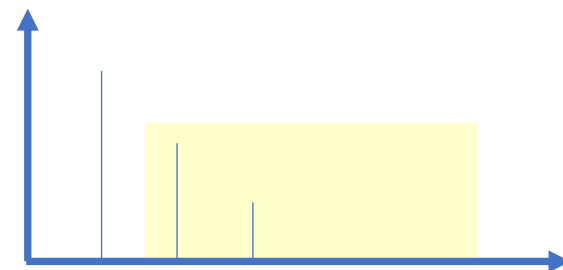
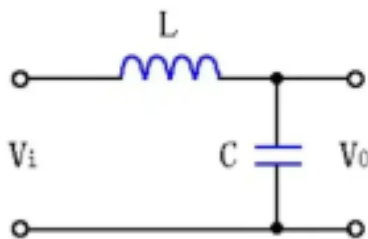
原始信号



正交基函数
(字典)



压缩，存储，表示



滤波，处理

回顾——周期信号的傅立叶级数

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➤ 复指数形式傅立叶级数

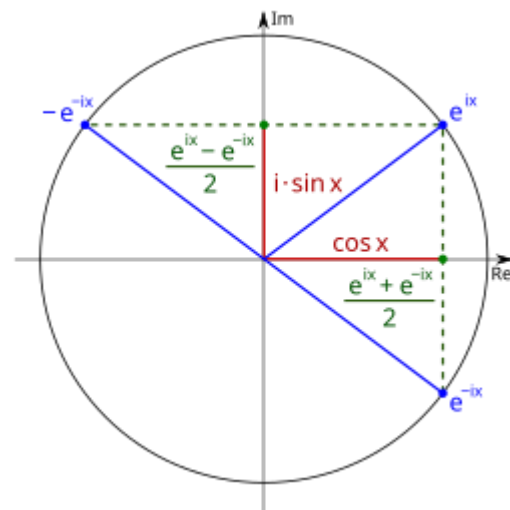
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$



$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$\cos n\omega_1 t = (e^{jn\omega_1 t} + e^{-jn\omega_1 t}) / 2$$

$$\sin n\omega_1 t = (e^{jn\omega_1 t} - e^{-jn\omega_1 t}) / (2j)$$



回顾——周期信号的傅立叶级数

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➤ 复指数形式傅立叶级数

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

欧拉公式：

$$\cos n\omega_1 t = (e^{jn\omega_1 t} + e^{-jn\omega_1 t}) / 2$$

$$\sin n\omega_1 t = (e^{jn\omega_1 t} - e^{-jn\omega_1 t}) / (2j)$$



$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[\frac{a_n - jb_n}{2} e^{jn\omega_1 t} + \frac{a_n + jb_n}{2} e^{-jn\omega_1 t} \right]$$

记作
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$$f(t) = F(0) + \sum_{n=1}^{\infty} [F(n\omega_1) e^{jn\omega_1 t} + F(-n\omega_1) e^{-jn\omega_1 t}]$$

简写为

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

，其中：

$$F_n = F(n\omega_1)$$

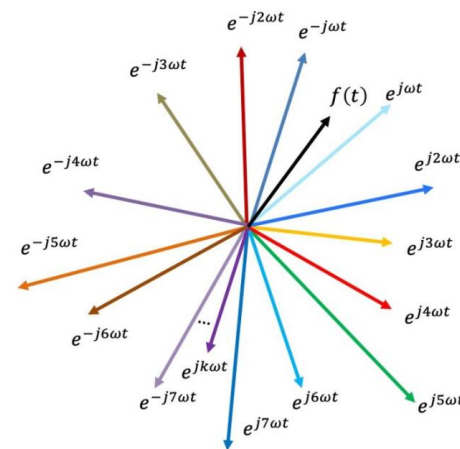
$$F_n = \frac{a_n - jb_n}{2} = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) (\cos n\omega_1 t - j \sin n\omega_1 t) dt$$

回顾——周期信号的傅立叶级数

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➤ 复指数形式傅立叶级数

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$



$$c_j = \frac{\langle f(t), \varphi_j(t) \rangle}{\langle \varphi_j(t), \varphi_j(t) \rangle} = \frac{1}{k_j} \langle f(t), \varphi_j(t) \rangle$$

$$k_i = \int_{t_0}^{t_0+T_1} e^{jn\omega_1 t} e^{-jn\omega_1 t} dt = T_1$$

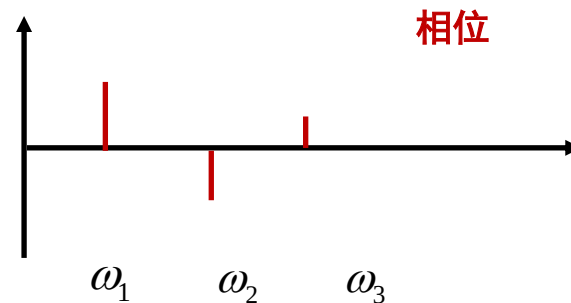
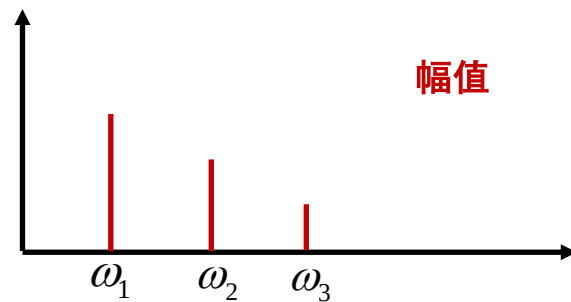
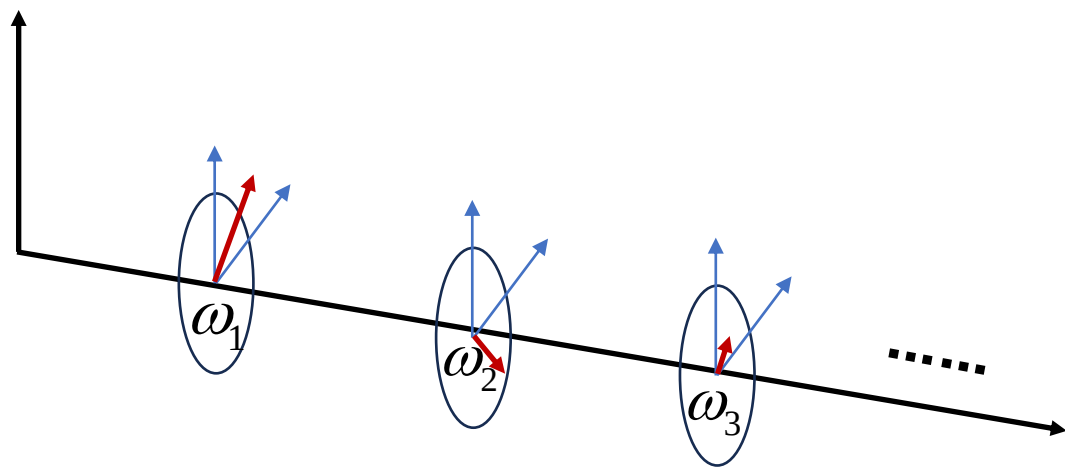
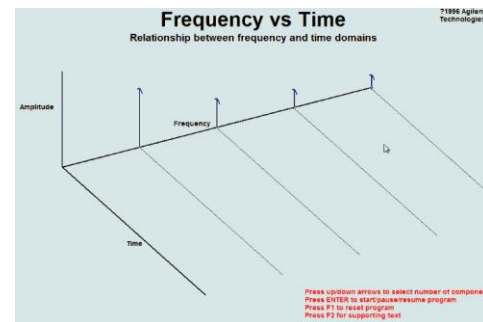
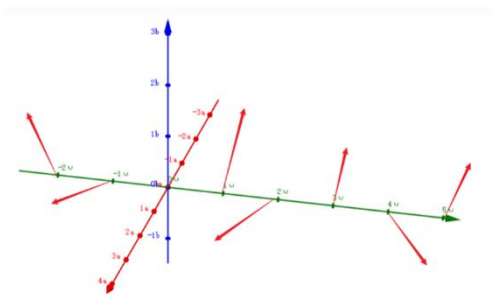
$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$$

回顾——周期信号的傅立叶级数

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➤ 复指数形式傅立叶级数

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$



回顾——周期信号的傅立叶级数

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➤ 复指数形式傅立叶级数

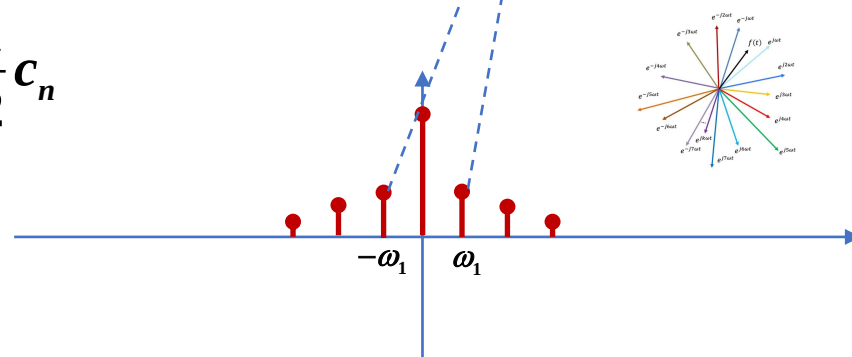
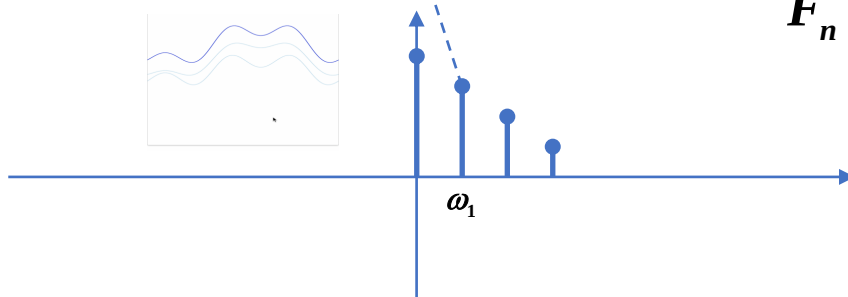
负频率的物理意义

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$
$$= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n)$$

Why?

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$$F_n = \frac{1}{2} c_n$$



为何实信号有负频率：sin或cos写成指数形式从欧拉公式的角度分成了 $e^{j\omega t}$ 和 $e^{-j\omega t}$ 两项，引入了负数部分，负频率的出现本身无物理意义，实信号分解后正负是成对出现的

回顾——周期信号的傅立叶级数

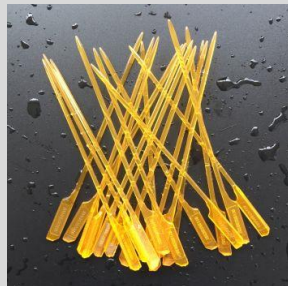
10

➤ 帕斯瓦尔定理：能量守恒

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{i=1}^{\infty} \|c_i\|^2 k_i$$

$$k_n = \int_{t_1}^{t_2} \varphi_n(t) \cdot \varphi_n^*(t) dt$$

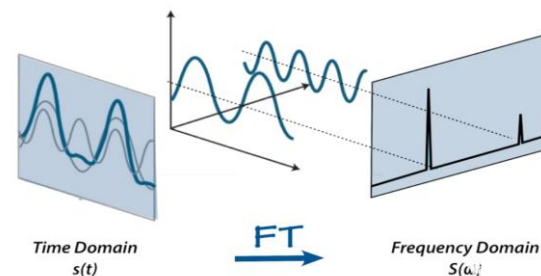


$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$$\int_{t_1}^{t_2} \|f(t)\|^2 dt = \sum_{n=-\infty}^{\infty} \|F_n\|^2 k_n$$

$$k_n = \int_{t_1}^{t_2} e^{jn\omega_1 t} \cdot e^{-jn\omega_1 t} dt$$

$$\int_{T_1} \|f(t)\|^2 dt = \sum_{n=-\infty}^{\infty} \|F_n\|^2 T_1$$



➤ 频谱

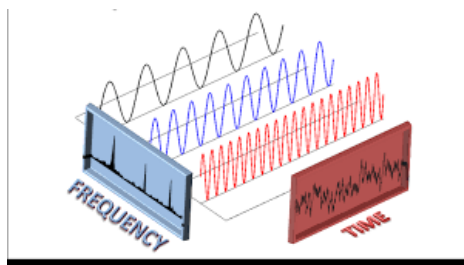
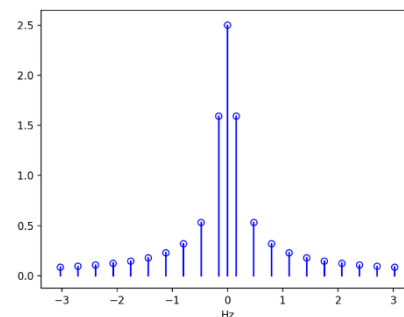
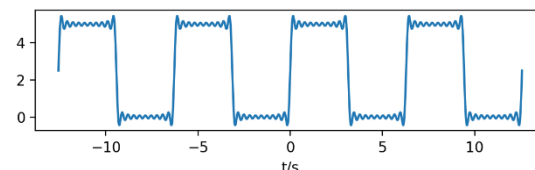
$$\{F_n\} \quad \text{FS谱} \quad \longrightarrow \quad |F_n| \quad \varphi_n = \text{Arg}(F_n)$$

FS幅度谱

FS相位谱

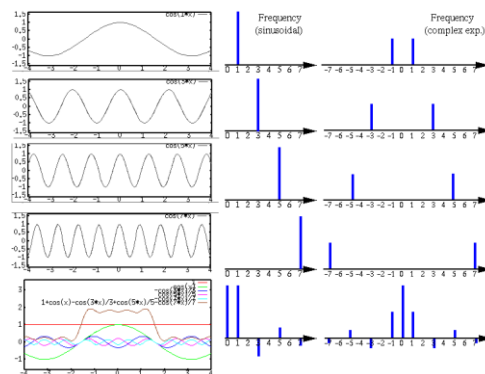
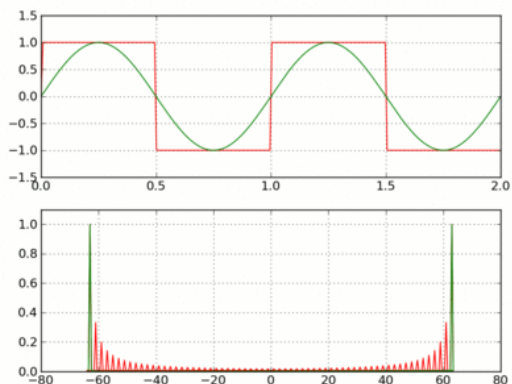
周期信号的傅立叶频谱特点：

- (1) 仅在一些离散频率点(nf_1) 上有值 $n\omega_1$
- (2) 离散间隔为 f_1 $\omega_1 = f_1 = 2\pi / T_1$
- (3) F_n 是双边谱，正负频率的频谱幅度相加才是实际幅度。
- (4) 信号的功率为 $\sum_{n=-\infty}^{\infty} |F_n|^2$ 由于复指数完备正交函数集中含有正负项，故为双边谱



周期信号的傅立叶级数——总结

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$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t)$$

$$\begin{cases} a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt \\ a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega_1 t dt \\ b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega_1 t dt \end{cases}$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

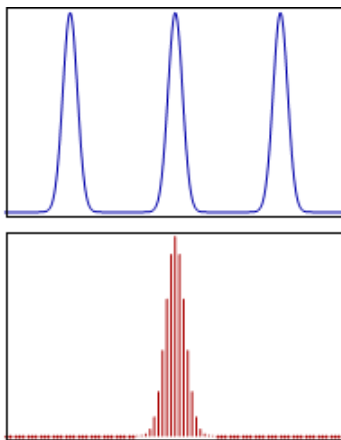
$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt$$

$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

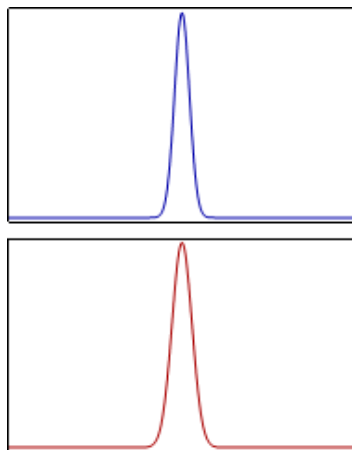
Why?



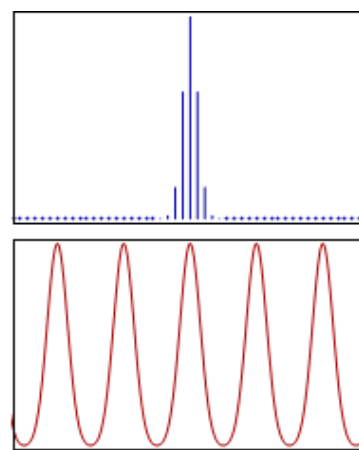
连续周期信号



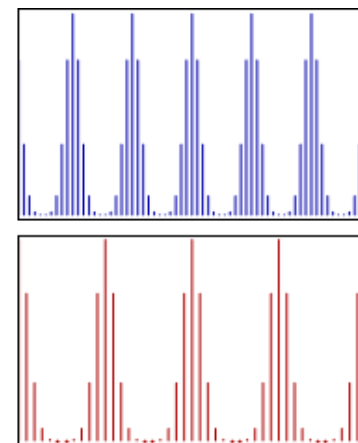
连续信号



离散信号

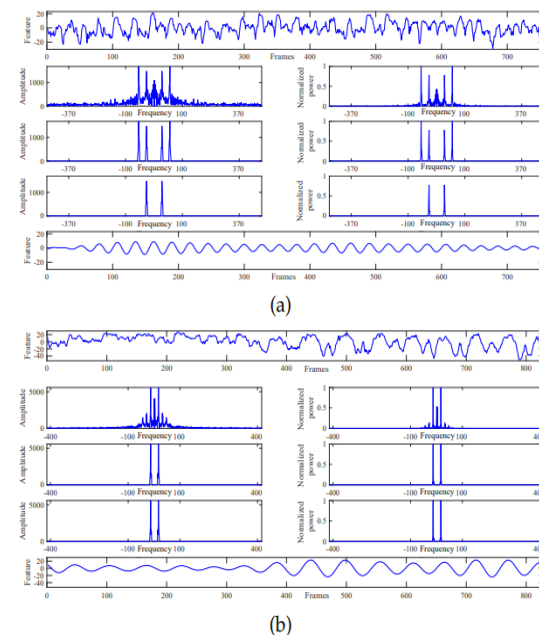
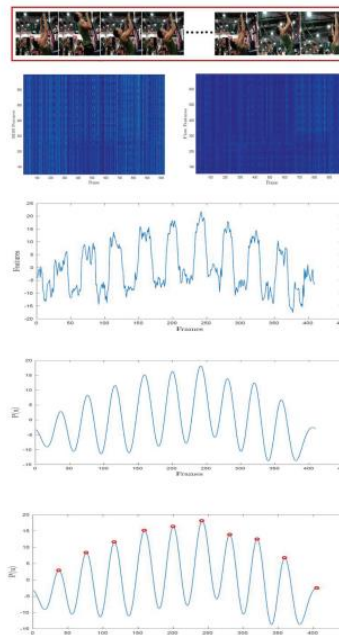
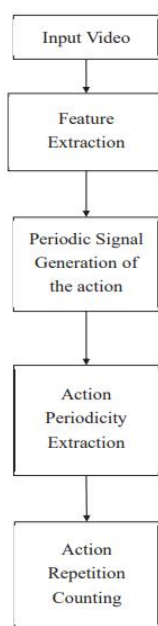
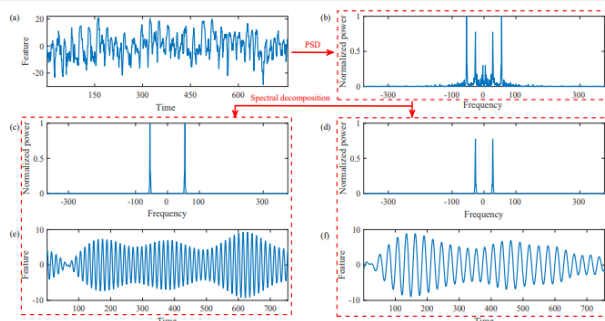
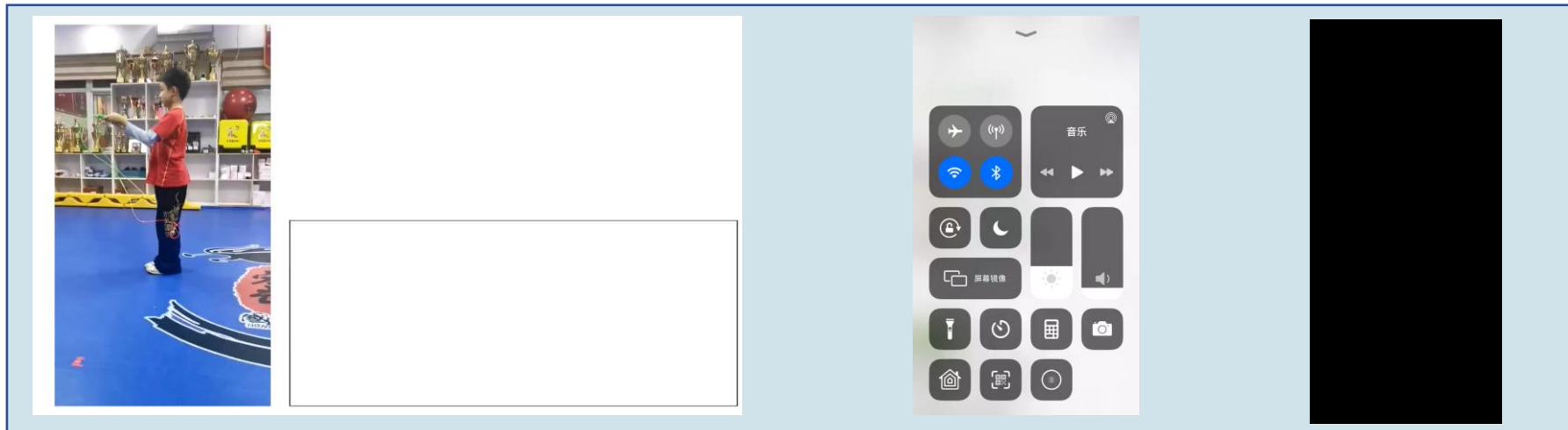


离散周期信号



$\{1, \cos n\omega_1 t, \sin n\omega_1 t\}$

$\{e^{jn\omega_1 t}\}$



Energy-based Periodicity Mining with Deep Features for Action Repetition Counting in Unconstrained Videos

Jianqin Yin, Yanchun Wu, Chaoran Zhu, Zijin Yin, Huaping Liu, Yonghao Dang, Zhiyi Liu, and Jun Liu

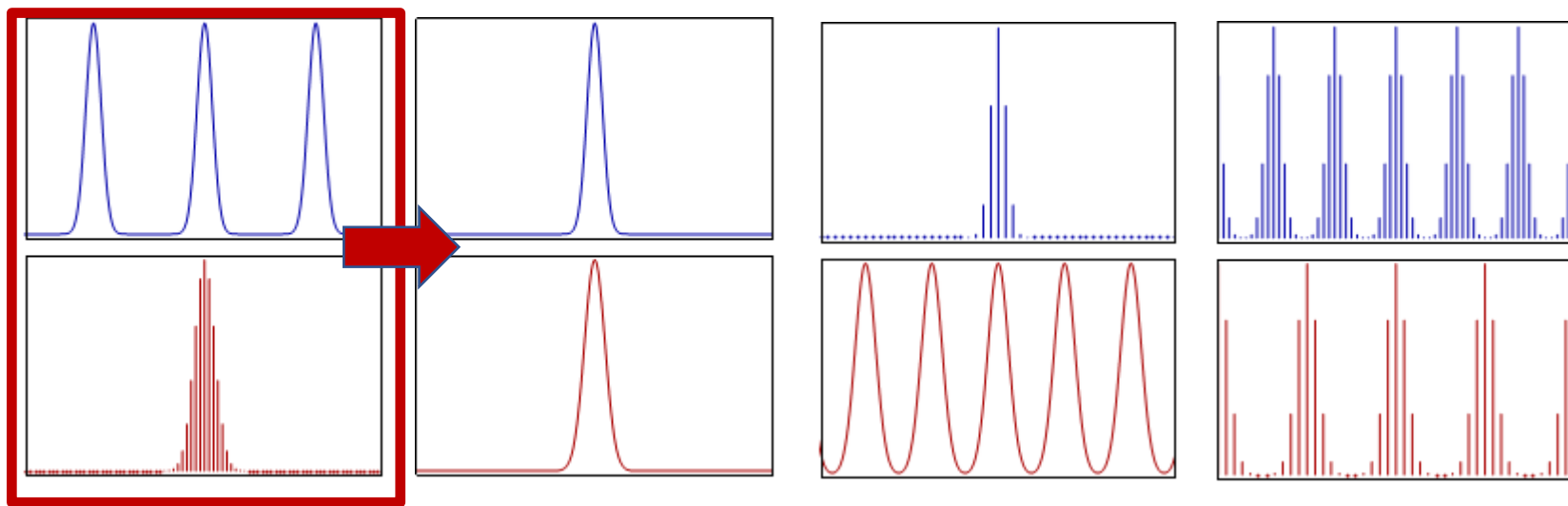
Abstract—Action repetition counting is to estimate the occurrence times of the repetitive motion in one action, which is a relatively new, significant, but challenging measurement problem. To solve this problem, we propose a new method superior to the traditional ways in two aspects, without preprocessing and applicable for arbitrary periodicity actions. Without preprocessing, the proposed model makes our scheme convenient for real applications; processing the arbitrary periodicity action makes our model more suitable for the actual circumstance. In terms of methodology, firstly, we extract action features using ConvNets and then propose an energy-based adaptive feature mode selection scheme to adaptively select proper deep feature mode according to the background of the video; secondly, we use Principal Component Analysis algorithm to generate the intuitive periodic information from the chaotic high-dimensional features; thirdly, we construct the periodic waveform of the action based on the high-energy rules by filtering the irrelevant information. Finally, we detect the peaks to obtain the times of the action repetition. Our work features two-fold: 1) We give a significant insight that features extracted by ConvNets for action recognition can well model the self-similarity periodicity of the repetitive action. 2) A high-energy based periodicity mining rule using features from ConvNets is presented, which can process arbitrary actions without preprocessing. Experimental results show that our method achieves superior or comparable performance on the three benchmark datasets, i.e. YT, Segments, QUA, and F4R5V.

Index Terms—Action repetition counting, Deep ConvNets

是结束还是开始

只能针对周期信号

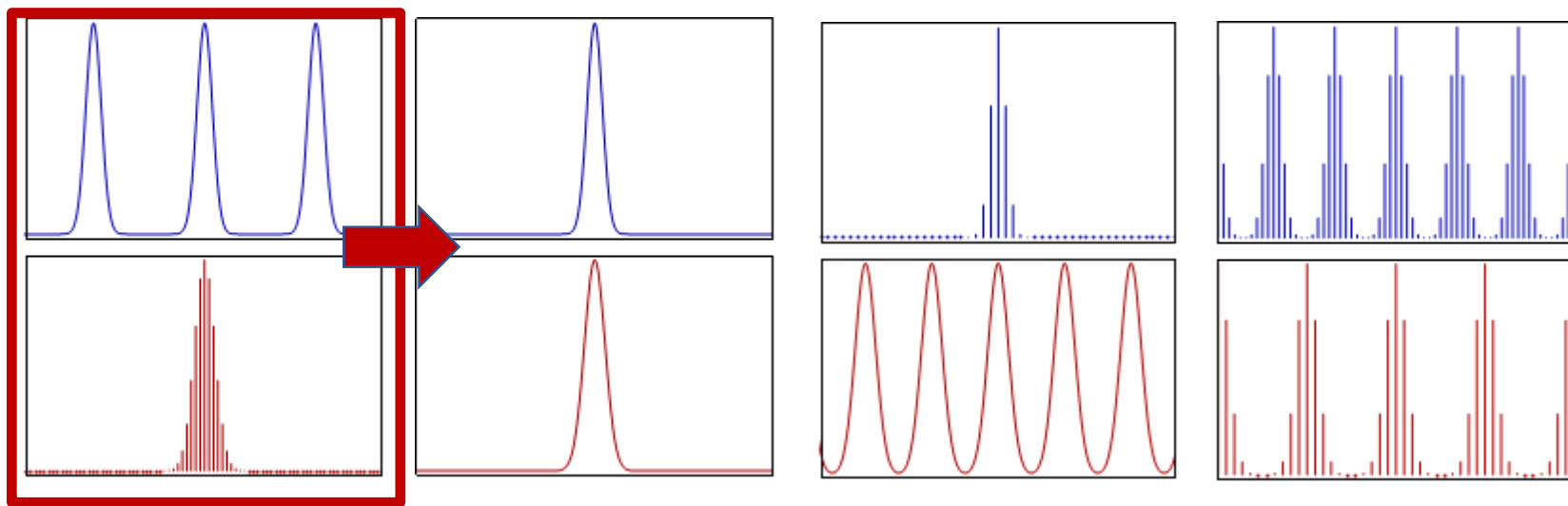
非周期信号怎么办



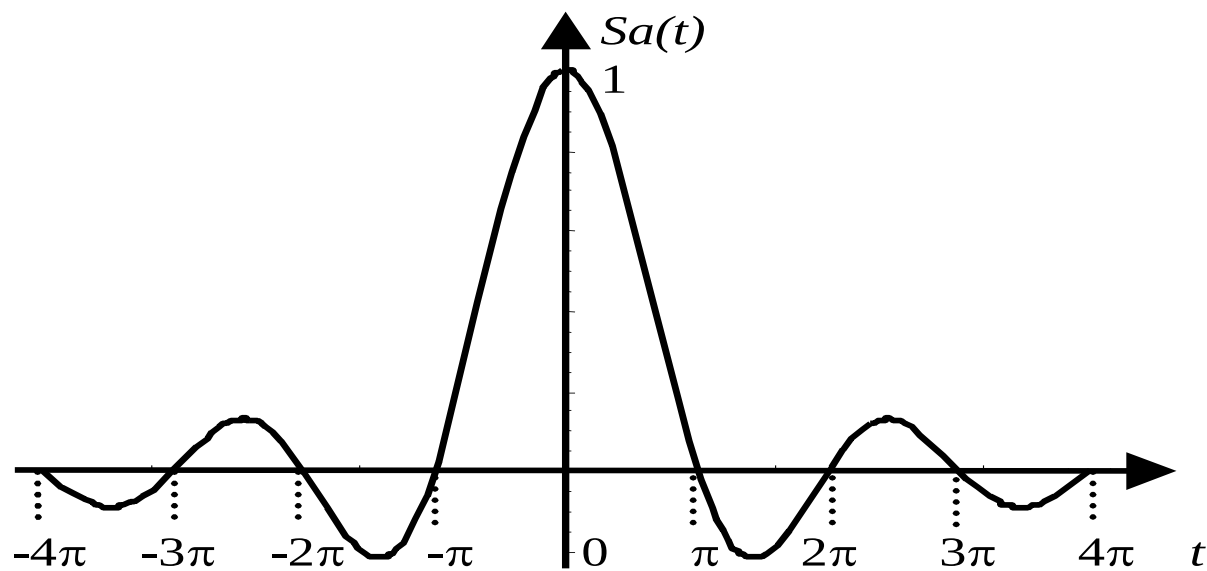
是结束还是开始

只能针对周期信号

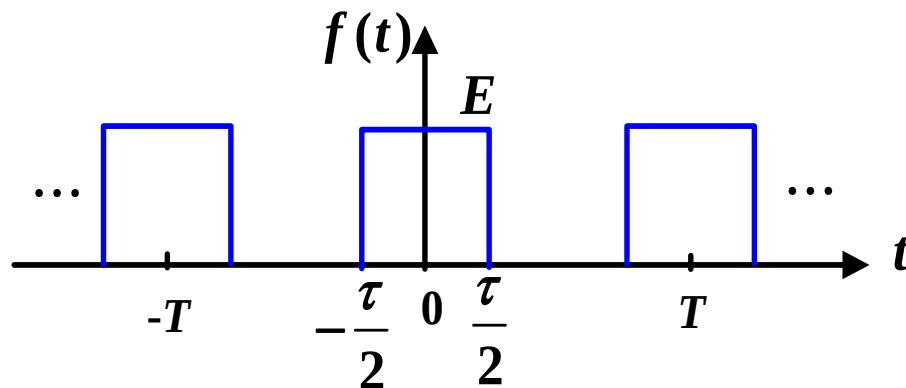
非周期信号怎么办



$$Sa(t) = \frac{\sin t}{t}$$



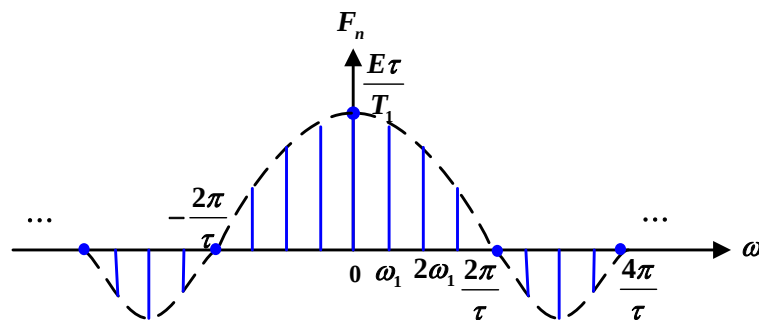
➤ 周期矩形脉冲信号



$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_1 t dt = \frac{2E}{T} \int_{-\tau/2}^{\tau/2} \cos n\omega_1 t dt = \frac{2E}{T} \frac{1}{n\omega_1} \sin n\omega_1 t \Big|_{-\tau/2}^{\tau/2} = \frac{2E}{T} \frac{1}{n\omega_1} \cdot 2 \sin\left(\frac{n\omega_1 \tau}{2}\right)$$

$$= \frac{2E}{n\pi} \sin\left(\frac{n\omega_1 \tau}{2}\right) = \frac{2E}{n\pi} \frac{\sin\left(\frac{n\omega_1 \tau}{2}\right)}{\frac{n\omega_1 \tau}{2}} \frac{n\omega_1 \tau}{2}$$

$$= \frac{2E\tau}{T} \text{Sa}\left(\frac{n\omega_1 \tau}{2}\right)$$

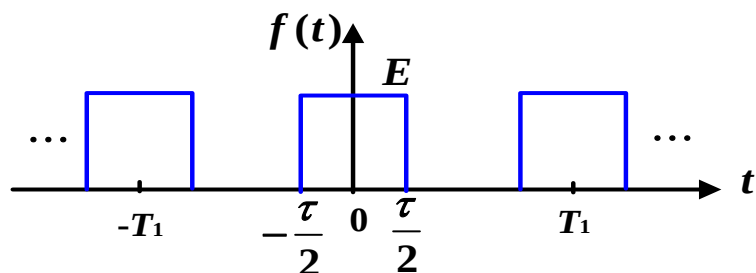


$$F_n = \frac{a_n - jb_n}{2}$$

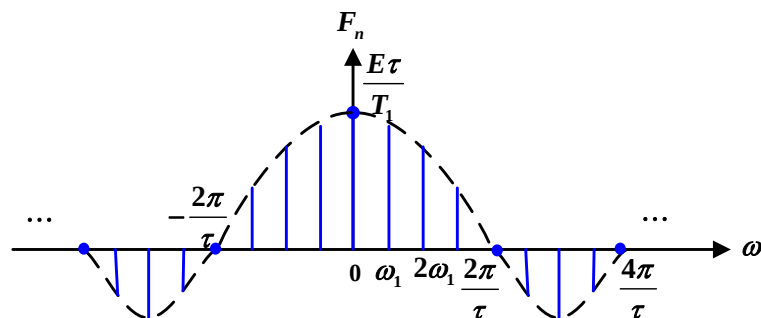
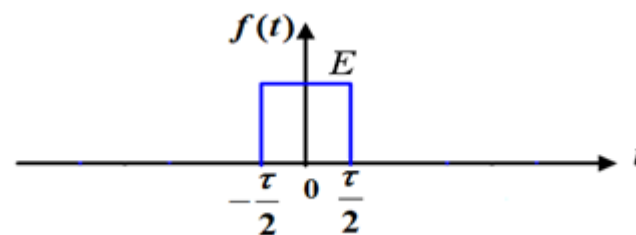
$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1 \tau}{2}\right) e^{jn\omega_1 t}$$

➤ 周期矩形脉冲信号

$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$



$T_1 \rightarrow \infty$



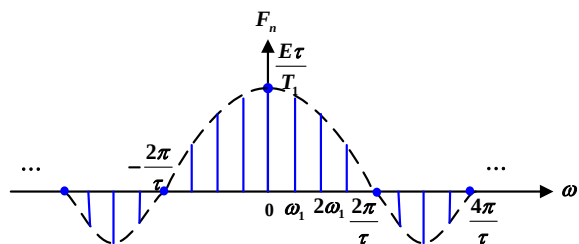
$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0$ 谱线间距变密直至为零
 ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零

非周期信号的傅里叶变换

➤ 周期 vs. 非周期的FS

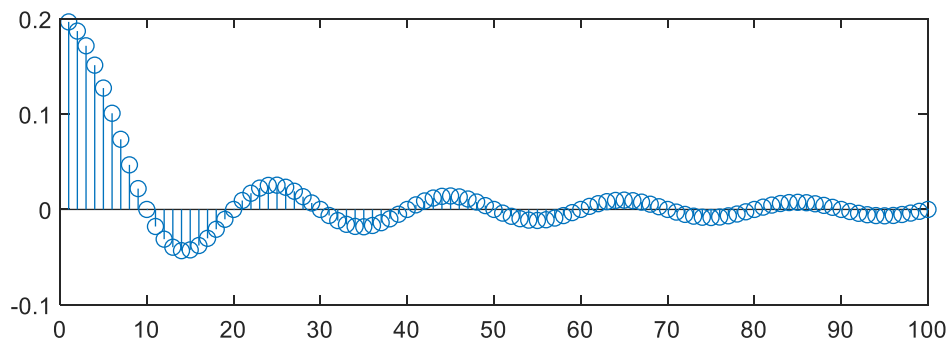
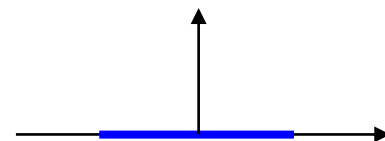


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

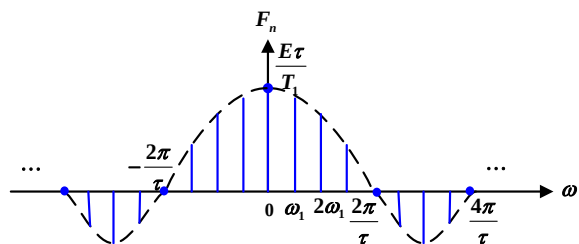
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

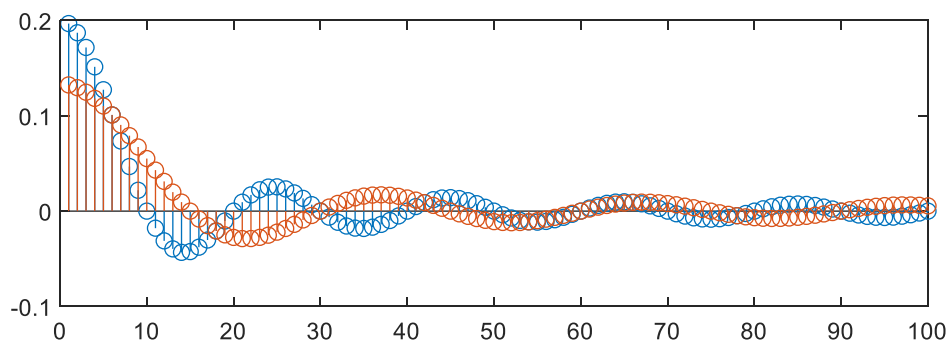
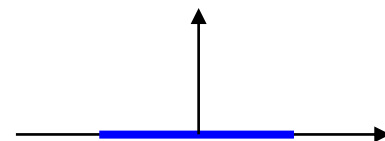


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

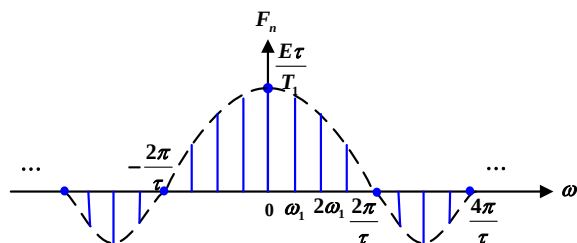
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

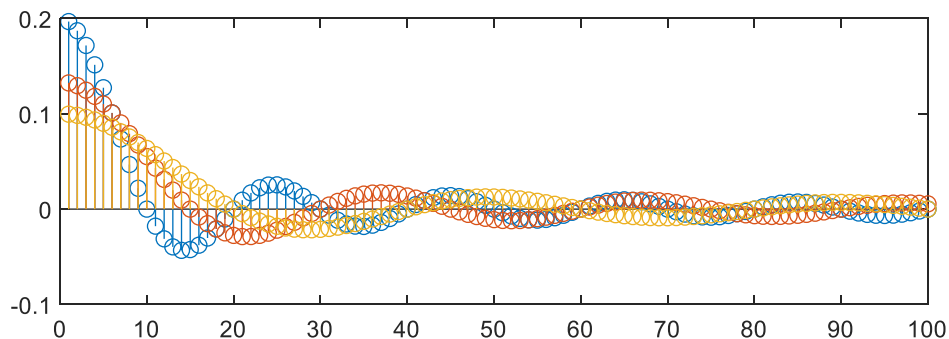
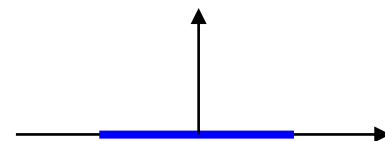


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

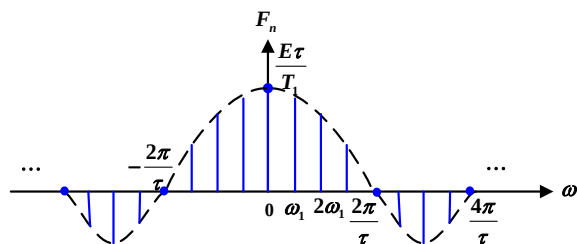
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

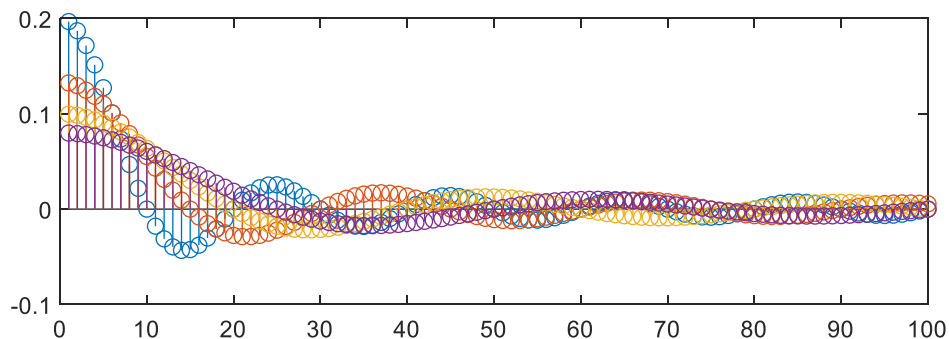
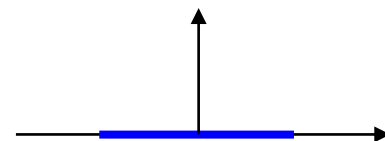


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

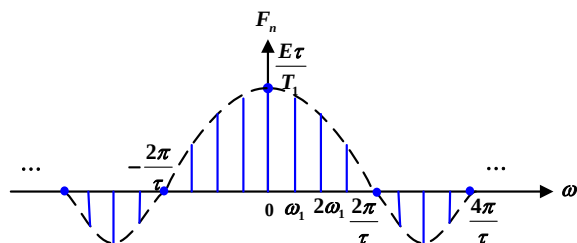
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$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

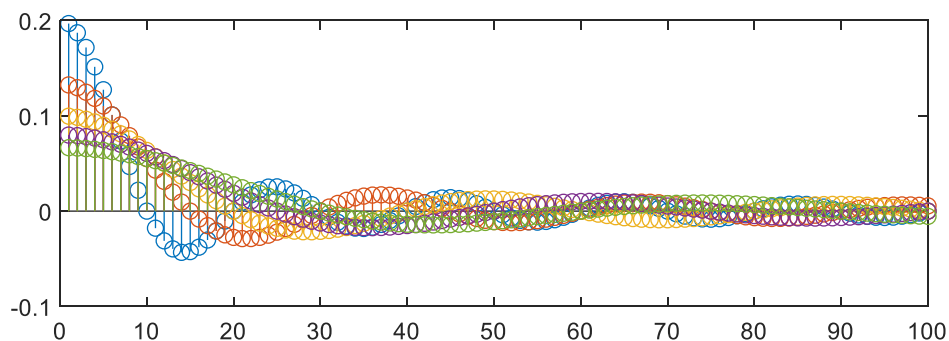
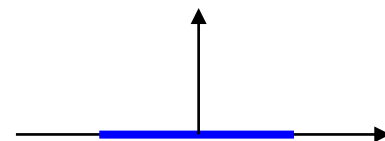


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

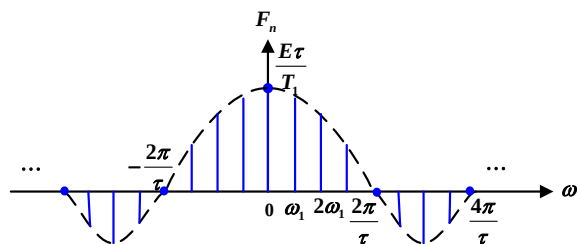
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

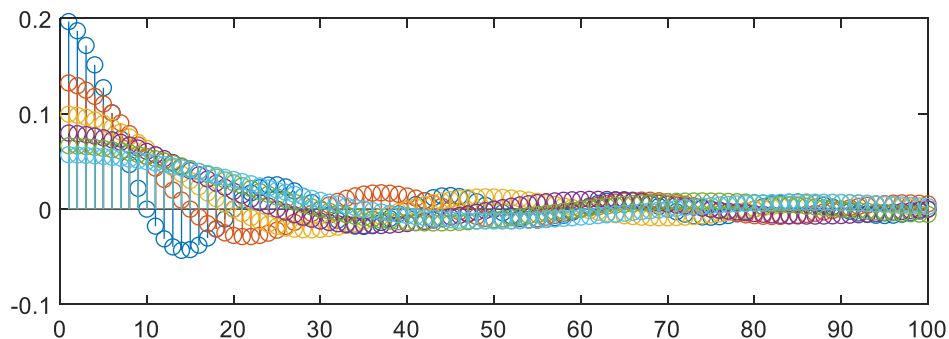
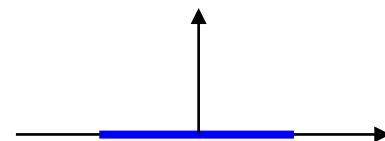


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

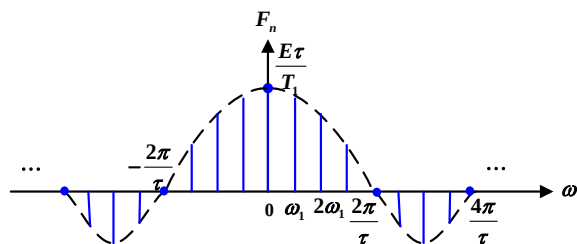
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

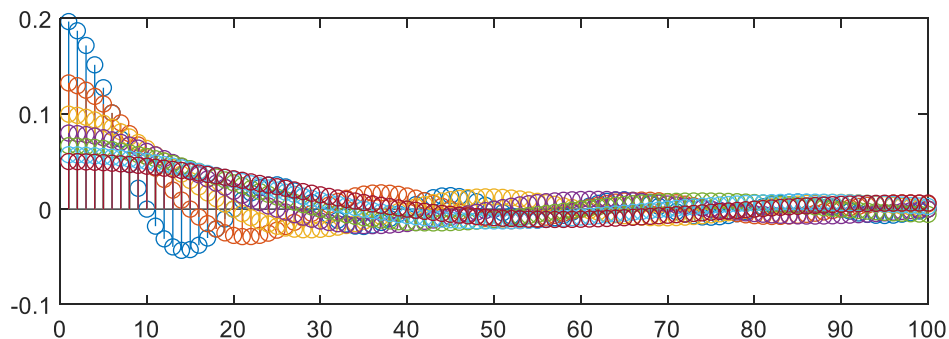
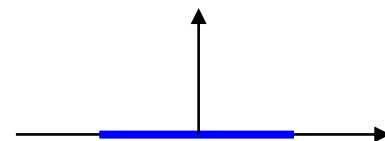


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

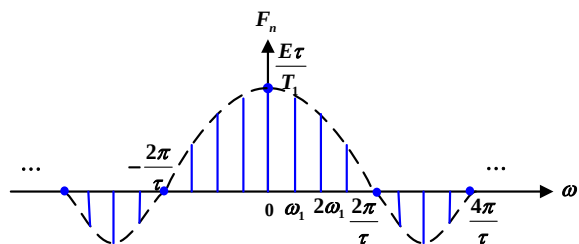
ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



➤ 周期 vs. 非周期的FS

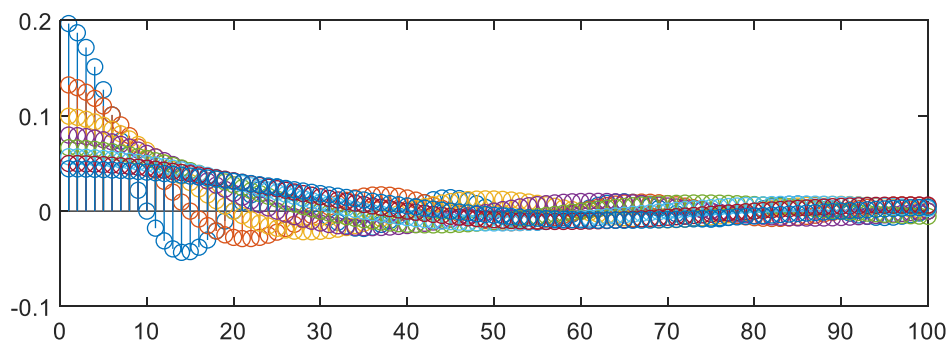
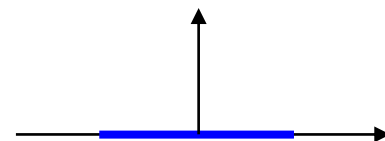


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

ω 变为连续域

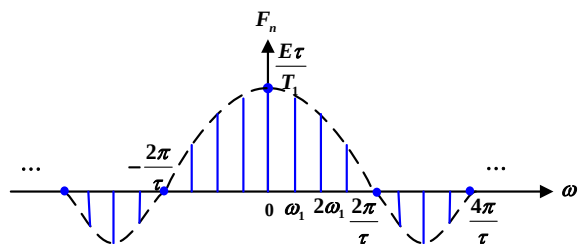
$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



失去意义！

➤ 周期 vs. 非周期的FS

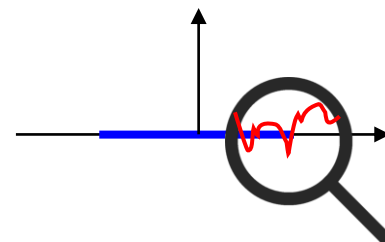


$$\omega_1 = \frac{2\pi}{T_1} \rightarrow 0 \quad \text{谱线间距变密直至为零}$$

ω 变为连续域

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

谱线高度变矮直至为零



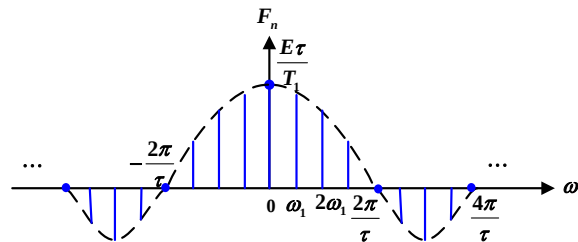
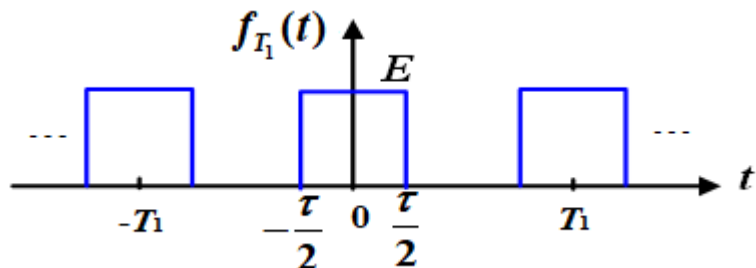
- 物理意义入手：既是信号，必有能量。因此，频域必会以某种形式**存在**
- 数学角度思考：无数个无穷小量之和，在极限意义下可能等于一个**有限值**。前述问题只是说每个分量变成无穷小量，但并不是说总和（信号）为零

$$F_n = \frac{1}{T_1} \int_{T_1} f(t) e^{-jn\omega_1 t} dt \rightarrow 0$$

$$0 \cdot \infty \rightarrow ?$$

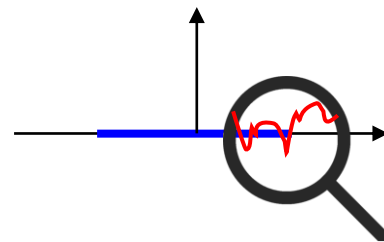
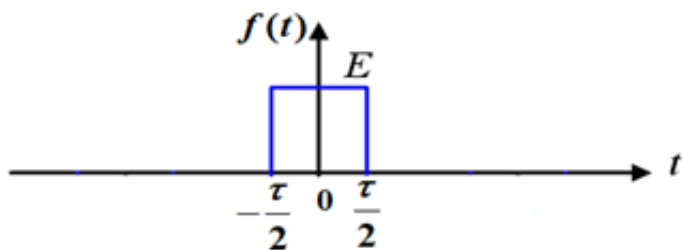
非周期信号的傅里叶变换

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$T_1 \rightarrow \infty$

$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$



$$f(t) = \lim_{T_1 \rightarrow \infty} f_{T_1}(t)$$

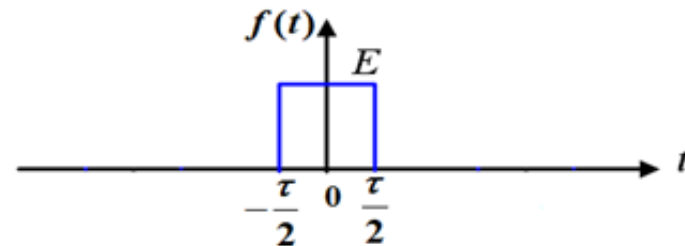
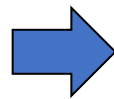
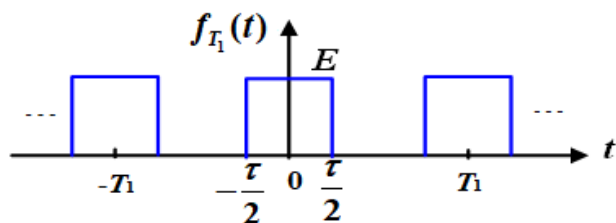
$\longrightarrow 0$

$$T_1 f_{T_1}(t) = \sum_{n=-\infty}^{\infty} E\tau \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$

密度

非周期信号的傅里叶变换

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$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

$$F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f_{T_1}(t) e^{-jn\omega_1 t} dt$$

$$\omega_1 = \frac{2\pi}{T_1}$$

$$T_1 \uparrow, \omega_1 \downarrow \rightarrow \Delta\omega$$

$$T_1 F_n = \int_{-T_1/2}^{T_1/2} f_{T_1}(t) e^{-jn\Delta\omega t} dt \stackrel{\text{记作}}{=} F(n\Delta\omega)$$

$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T_1} F(n\Delta\omega) e^{jn\Delta\omega t} = \sum_{n=-\infty}^{\infty} \frac{1}{2\pi} F(n\Delta\omega) e^{jn\Delta\omega t} \cdot \Delta\omega$$

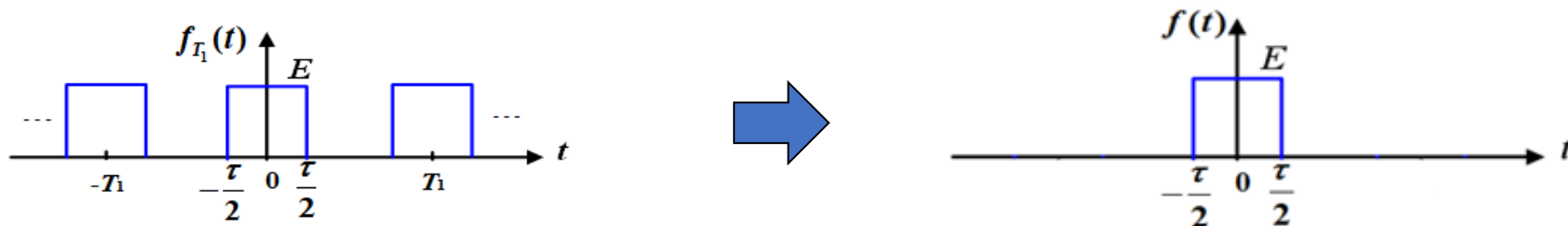
$$T_1 \rightarrow \infty$$

$$\lim_{T_1 \rightarrow \infty} f_{T_1}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

非周期信号的傅里叶变换

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$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$T_1 F_n = \int_{-T_1/2}^{T_1/2} f_{T_1}(t) e^{-jn\Delta\omega_1 t} dt \quad \text{记作} \quad F(n\Delta\omega)$$

$$\lim_{\Delta\omega \rightarrow 0} F(n\Delta\omega) = \lim_{T_1 \rightarrow \infty} \int_{-T_1/2}^{T_1/2} f_{T_1}(t) e^{-jn\Delta\omega t} dt$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \quad \text{记作} \quad = F(\omega)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \stackrel{\Delta}{=} \mathfrak{F}^{-1}[F(\omega)]$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \stackrel{\Delta}{=} \mathfrak{F}[f(t)]$$

$$f(t) \leftrightarrow F(\omega)$$

$$F(\omega) = \lim_{T_1 \rightarrow \infty} T_1 F_n = \lim_{\Delta\omega \rightarrow 0} \frac{2\pi}{\Delta\omega} F_n = \lim_{\Delta f \rightarrow 0} \frac{F_n}{\Delta f}$$

$F(\omega)$ 称为频谱密度，简称**频谱**

FT存在的充分条件：绝对可积 $\int_{-\infty}^{+\infty} |f(t)| dt < \infty$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F(\omega) = |F(\omega)| e^{j\angle F(\omega)}$$

$$|F(\omega)| \sim \omega$$

幅度频谱反映各分量相对大小的关系

$$\angle F(\omega) \sim \omega$$

相位频谱反映各分量的初始相位

$$\begin{aligned} F(\omega) &= R(\omega) + jX(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) \cos \omega t dt - j \int_{-\infty}^{\infty} f(t) \sin \omega t dt \end{aligned}$$

$$R(\omega) = \int_{-\infty}^{\infty} f(t) \cos \omega t dt \quad X(\omega) = \int_{-\infty}^{\infty} f(t) \sin \omega t dt$$

$$R(\omega) = R(-\omega)$$

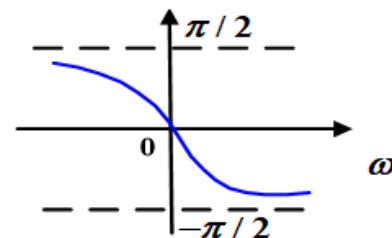
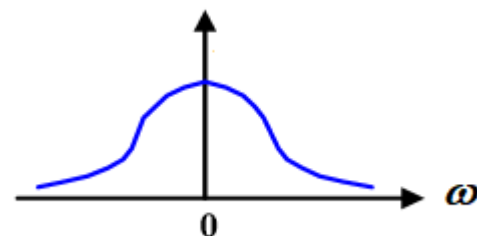
频谱实部是偶对称的

$$X(\omega) = -X(-\omega)$$

频谱虚部是奇对称的

$$\varphi(\omega) = \arctan \frac{X(\omega)}{R(\omega)}$$

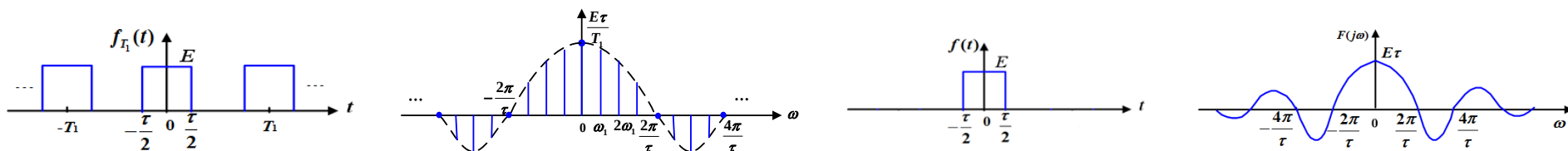
频谱相位是奇对称的



非周期信号的傅里叶变换

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$F(\omega)$ 与 F_n (周期信号频谱) 的区别:



$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f_{T_1}(t) e^{-jn\omega_1 t} dt$$

$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}, \omega_1 = \frac{2\pi}{T_1}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

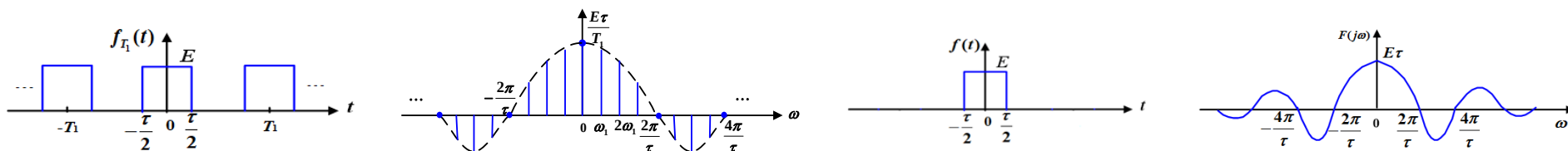
F_n 是离散的, 代表各分量**绝对大小**

$F(\omega)$ 是连续的, 代表各分量**相对大小**

非周期信号的傅里叶变换

37

$F(\omega)$ 与 F_n (周期信号频谱) 的区别:



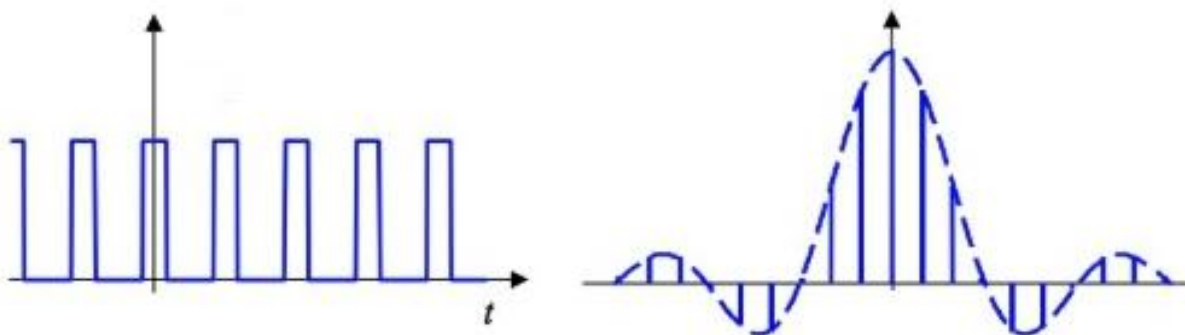
$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f_{T_1}(t) e^{-jn\omega_1 t} dt$$
$$f_{T_1}(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}, \omega_1 = \frac{2\pi}{T_1}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

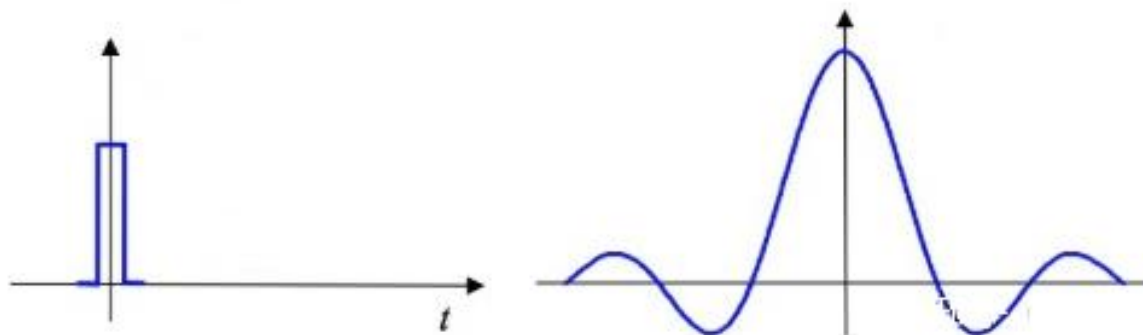
	FS	FT
被分析对象	周期信号	非周期信号
频率定义域	离散频率, 谐波频率处	连续频率, 整个频率轴
函数值意义	频率分量的数值	频率分量的密度值

非周期信号的傅里叶变换

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$$F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f_{T_1}(t) e^{-jn\omega_1 t} dt$$

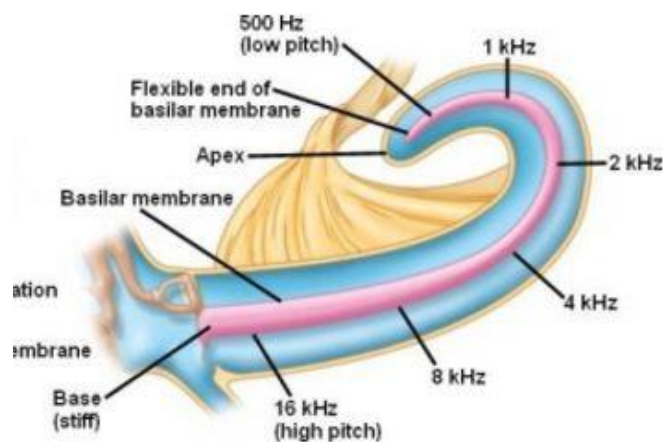
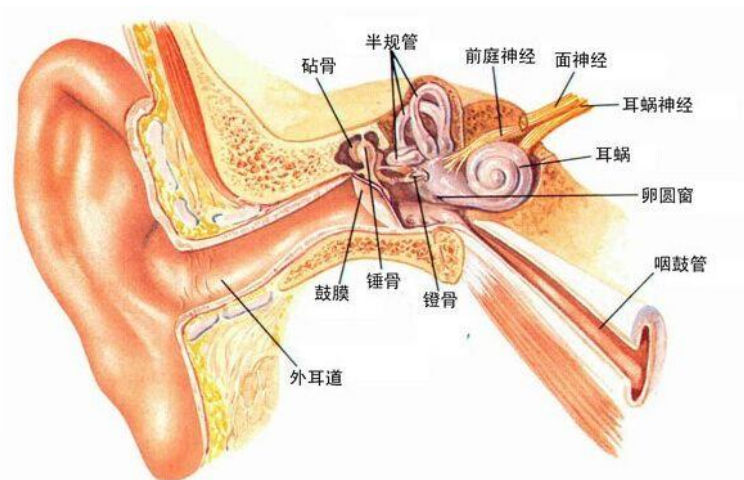


$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$T_1 F_n$$

非周期信号的傅里叶变换

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非周期信号的傅里叶变换

常用信号举例

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega$$

➤ 常用信号的FT

例1: $f(t) = e^{-at}u(t) \quad (a > 0)$ 求 $F(\omega)$

解:

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \left. \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right|_0^{\infty} \\ &= \frac{1}{j\omega + a} \end{aligned}$$

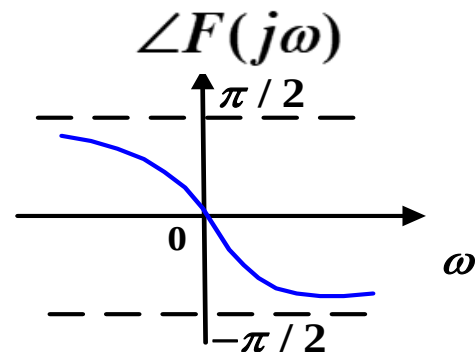
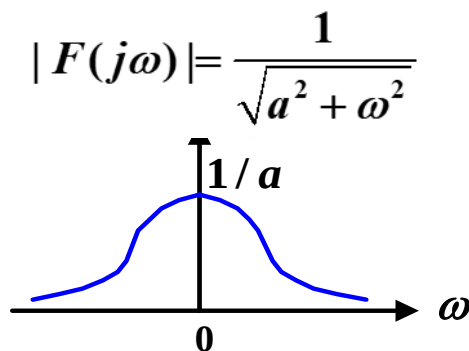
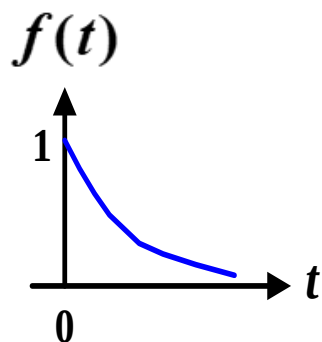
$$e^{-at}u(t) \leftrightarrow \frac{1}{j\omega + a}, \quad a > 0$$

➤ 常用信号的FT

例1: $f(t) = e^{-at}u(t) \quad (a > 0)$ 求 $F(\omega)$

解::
$$e^{-at}u(t) \leftrightarrow \frac{1}{j\omega + a}, \quad a > 0$$

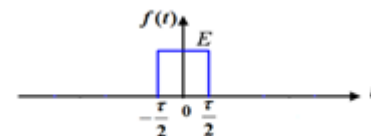
$$|F(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}} \quad \angle F(\omega) = -\arctan \frac{\omega}{a}$$



➤ 常用信号的FT

例2: 求矩形脉冲信号的频谱

$$f(t) = EG_{\tau}(t) \quad \text{其中} G_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & \text{其它} \end{cases}$$



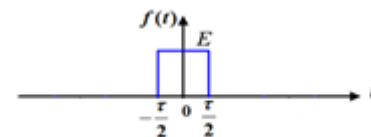
解:

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \\ &= \int_{-\tau/2}^{\tau/2} Ee^{-j\omega t} dt = E \frac{e^{-j\omega t}}{-j\omega} \bigg|_{-\tau/2}^{\tau/2} = E \frac{e^{-j\omega\tau/2} - e^{j\omega\tau/2}}{-j\omega} \\ &= E \frac{-2j \sin\left(\frac{\omega\tau}{2}\right)}{-j\omega} = E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right) \end{aligned}$$

➤ 常用信号的FT

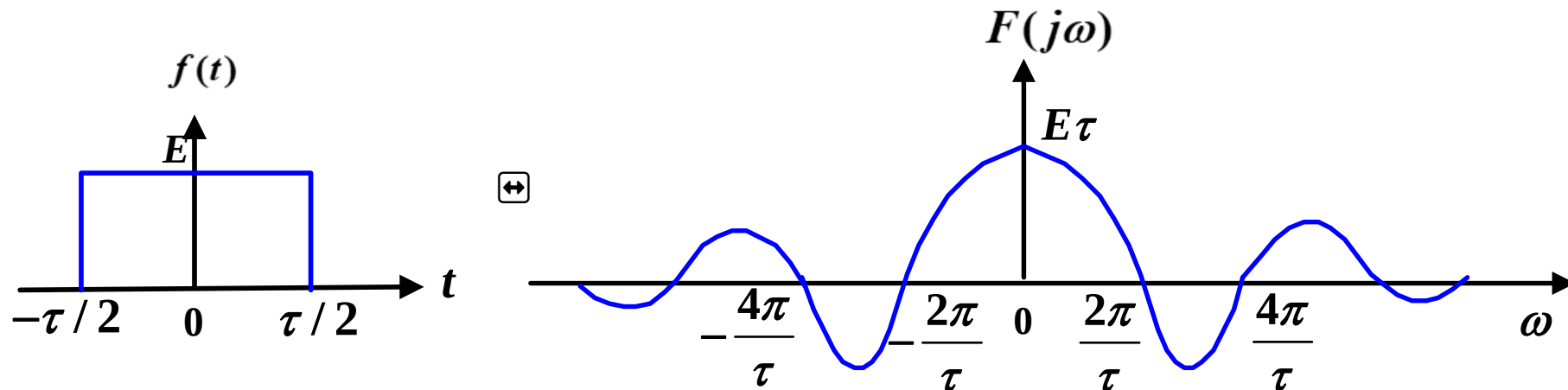
例2: 求矩形脉冲信号的频谱

$$f(t) = EG_{\tau}(t) \quad \text{其中} G_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & \text{其它} \end{cases}$$



解::

$$EG_{\tau}(t) \leftrightarrow E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

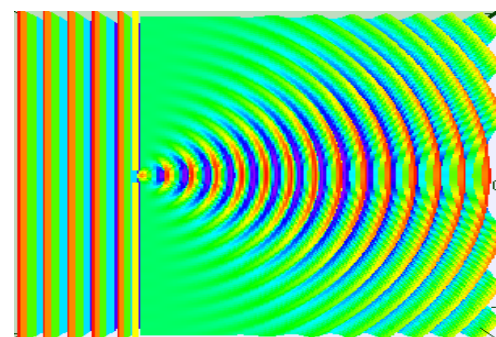
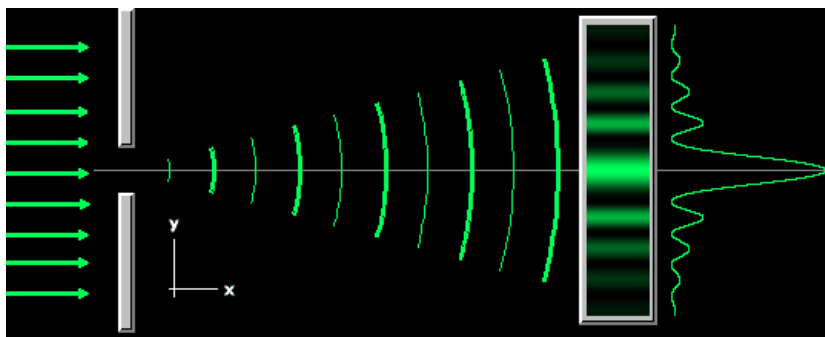
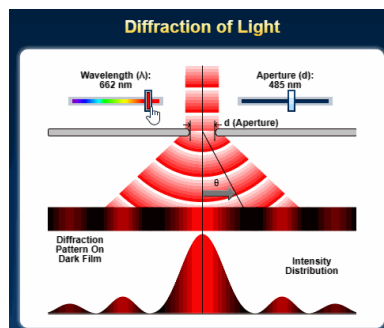
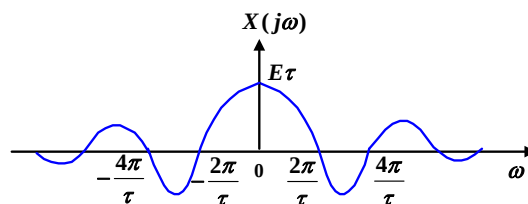
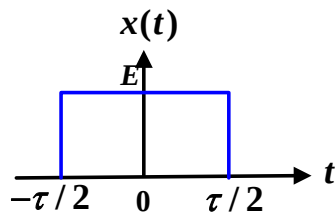
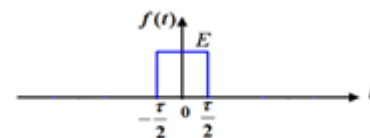


非周期信号的傅里叶变换

➤ 常用信号的FT

例2: 求矩形脉冲信号的频谱

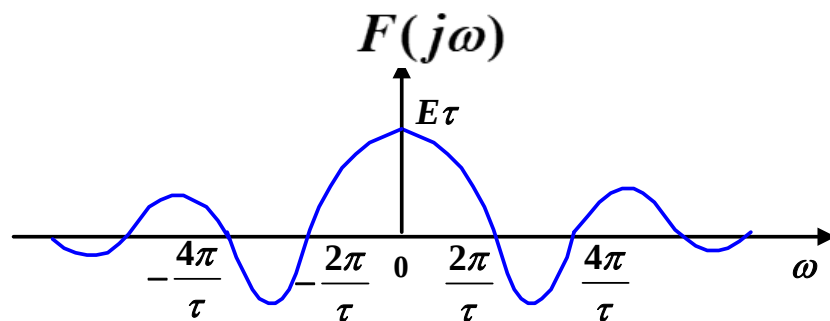
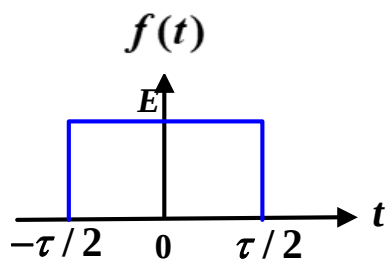
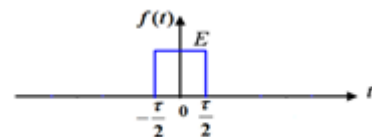
$$f(t) = EG_{\tau}(t) \quad \text{其中} G_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & \text{其它} \end{cases}$$



➤ 常用信号的FT

例2: 求矩形脉冲信号的频谱

$$f(t) = EG_{\tau}(t) \quad \text{其中} G_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & \text{其它} \end{cases}$$



矩形脉冲信号FT的特点:

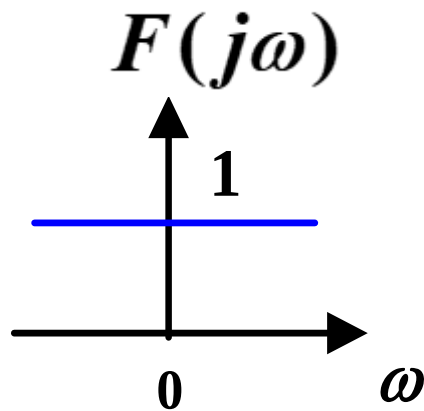
- FT为Sa函数, 原点处函数值等于矩形脉冲的面积
- FT的过零点位置为 $\omega = 2k\pi / \tau (k \neq 0)$
- 频域的能量集中在第一个过零点区间 $\omega \in (-2\pi / \tau, 2\pi / \tau)$
- 带宽只与脉宽有关, 与脉高E无关。带宽为 $B_{\omega} = 2\pi / \tau$

➤ 常用信号的FT

例3：求单位冲激信号 $\delta(t)$ 的频谱，并写出它的频域分解式

解：

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt \\ &= 1 \end{aligned}$$



$$\underline{\delta(t) \leftrightarrow 1}$$

频域分解式： $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$

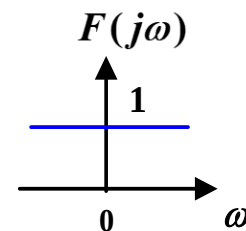
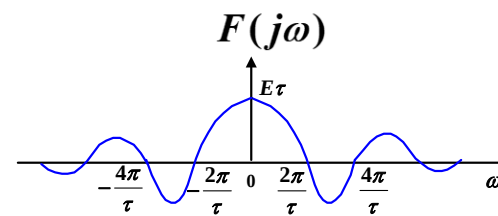
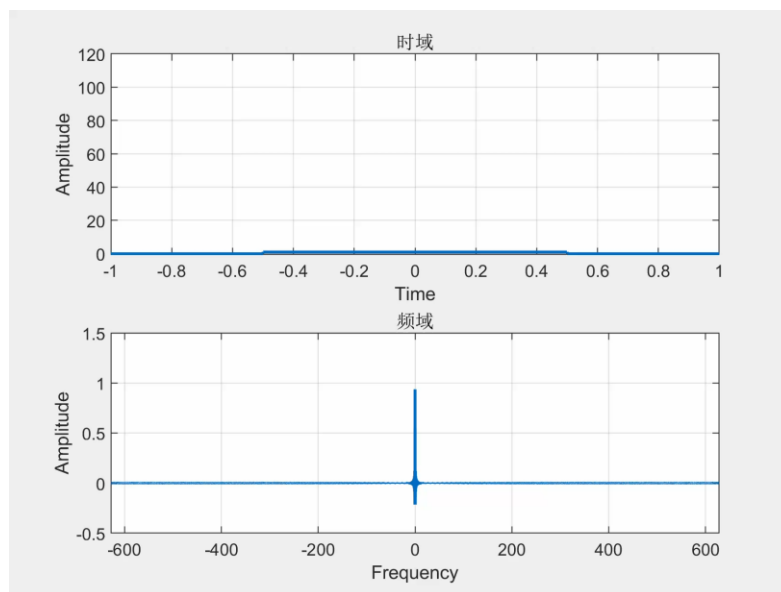
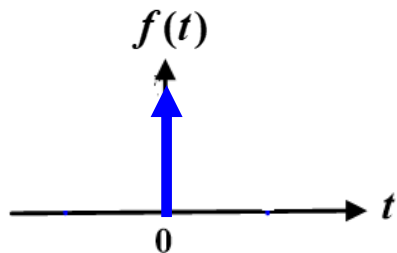
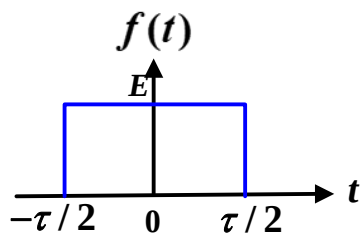
$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega$$

非周期信号的傅里叶变换

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➤ 常用信号的FT

例3：求单位冲激信号 $\delta(t)$ 的频谱，并写出它的频域分解式



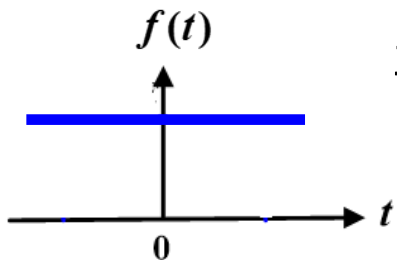
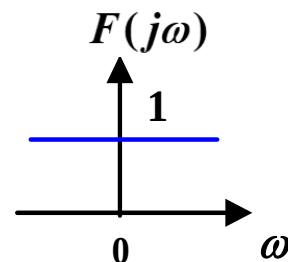
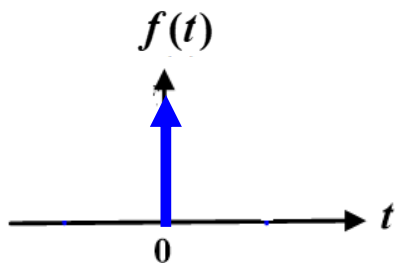
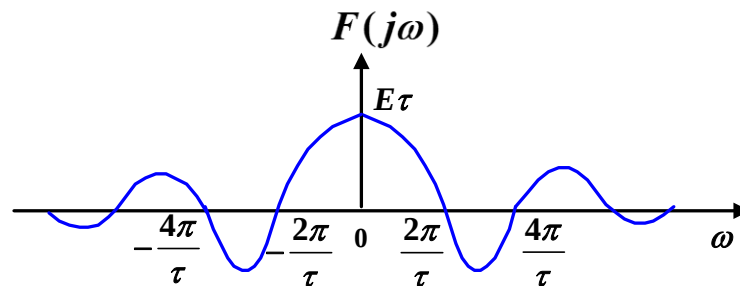
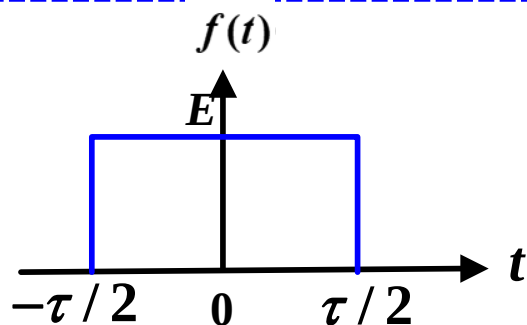
- 冲激函数的频谱等于**常数**，即在整个频率范围内频谱是均匀分布的。
- 说明在**时域中变化异常剧烈**的冲激函数中包含了**幅度相等的所有频率分布**。
- 因此，这种频谱常被称为均匀谱，或白色谱。

非周期信号的傅里叶变换

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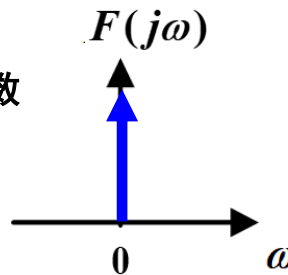
➤ 常用信号的FT

例3：求单位冲激信号 $\delta(t)$ 的频谱，并写出它的频域分解式



直流信号的傅里叶频谱是位于零点的冲激函数

$$F\left[\frac{1}{2\pi}\right] = \delta(\omega) \quad F[1] = 2\pi\delta(\omega)$$



➤ 常用信号的FT

$$e^{-at}u(t) \leftrightarrow \frac{1}{j\omega + a}, \quad a > 0$$

$$G_{\tau}(t) \leftrightarrow \tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$\delta(t) \leftrightarrow 1$$

$$u(t) \leftrightarrow \pi\delta(\omega) + \frac{1}{j\omega}$$

FT存在的充分条件：绝对可积

$$\int_{-\infty}^{+\infty} |f(t)| dt < \infty$$

傅里叶变换的性质

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

1 唯一性

2 线性特性

3 奇偶特性

4 对称特性

5 时频展缩特性

6 时移特性

7 频移特性

8 时域微分特性

9 频域微分特性

10 时域卷积定理

11 频域卷积定理

12 信号能量与频谱的关系

➤ FT的唯一性

若 $\mathcal{F}[f_1(t)] = \mathcal{F}[f_2(t)] = F(\omega)$ 则 $f_1(t) = f_2(t)$

反之 若 $\mathcal{F}^{-1}[F_1(\omega)] = \mathcal{F}^{-1}[F_2(\omega)] = f(t)$,
则 $F_1(\omega) = F_2(\omega)$

➤ FT是线性运算

对一个信号求FT，等于对其分量（信号）求FT然后再组合

线性

齐次性	$F[af(t)] = aF[f(t)]$
叠加性	$F[f_1(t) + f_2(t)] = F[f_1(t)] + F[f_2(t)]$

➡

$$F\left[\sum_n a_n f_n(t)\right] = \sum_n a_n F[f_n(t)]$$

➤ FT的反褶与共轭性

$$f(t) \Leftrightarrow F(\omega)$$

反褶与共轭	时域	频域
反褶	$f(-t)$	$F(-\omega)$
共轭	$f^*(t)$	$F^*(-\omega)$
反褶&&共轭	$f^*(-t)$	$F^*(\omega)$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

➤ FT的奇偶性

偶信号的频谱是偶函数，奇信号的频谱是奇函数。

证明： 若 $f(t)$ 是偶函数，即 $f(-t) = f(t)$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\begin{aligned} F(-\omega) &= \int_{-\infty}^{\infty} f(t) e^{j\omega t} dt \quad \text{令 } t = -\tau \\ &= \int_{-\infty}^{\infty} f(-\tau) e^{-j\omega \tau} d\tau = \int_{-\infty}^{\infty} f(\tau) e^{-j\omega \tau} d\tau \\ &= F(\omega) \end{aligned}$$

➤ FT与IFT的对偶性

- 极相似的公式背后隐藏着什么关系？
- 求IFT与FT本质上是相通的？

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega$$

➤ FT与IFT的对偶性

I: FT与IFT的变换核函数是共轭对称的

$$\{e^{-j\omega t}\}^* = e^{j\omega t} \quad \{e^{j\omega t}\}^* = e^{-j\omega t}$$

$$\mathcal{F}^{-1}[F(\omega)] = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} (F^*(\omega) e^{-j\omega t} d\omega)^*$$

$$= \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} (F^*(\omega) e^{-j\omega t} d\omega) \right)^* = \frac{1}{2\pi} \left\{ \mathcal{F}_{\omega} [F^*(\omega)] \right\}^*$$

其中, $\mathcal{F}_{\omega} [F^*(\omega)]$ 表示按自变量 ω 进行FT, 结果仍是 t 的函数。



在计算机程序设计实现上, IFT可以通过FT来完成

➤ FT与IFT的对偶性

II: $F(t) \Leftrightarrow 2\pi f(-\omega)$

证明: $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad f(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-j\omega t} d\omega$

将变量 t 与 ω 互换，可以得到：

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t) e^{-j\omega t} dt$$

等号右边是对函数 $F(t)$ 的傅里叶变换！

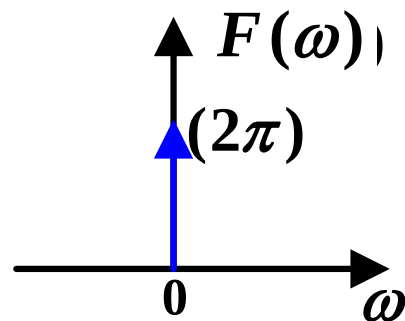
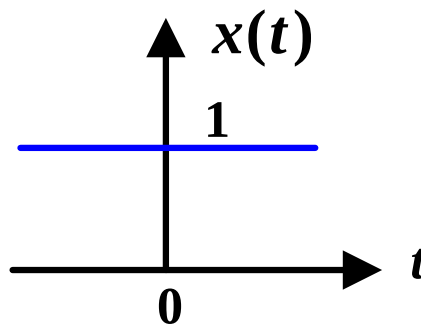
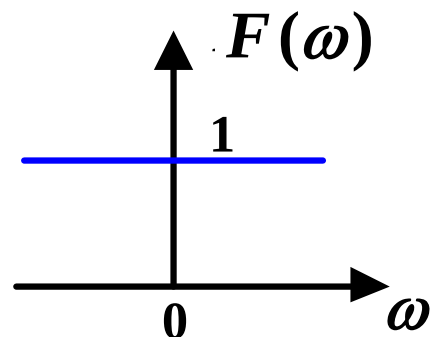
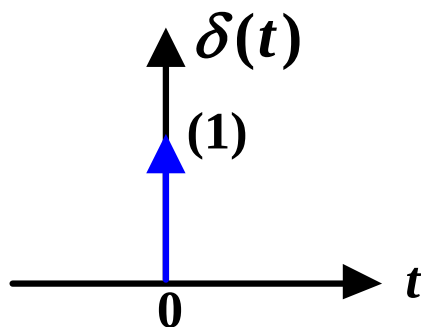
➤ FT与IFT的对偶性

II: $F(t) \Leftrightarrow 2\pi f(-\omega)$

$$\delta(t) \Leftrightarrow 1$$

$$1 \Leftrightarrow 2\pi\delta(-\omega)$$

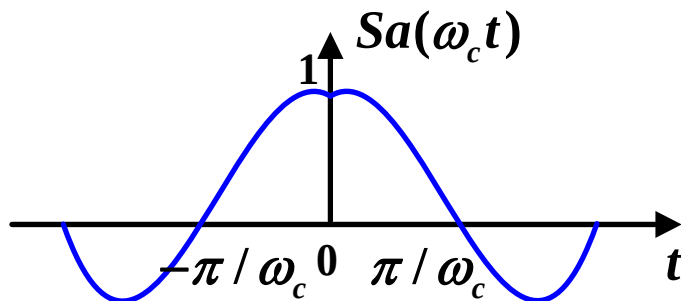
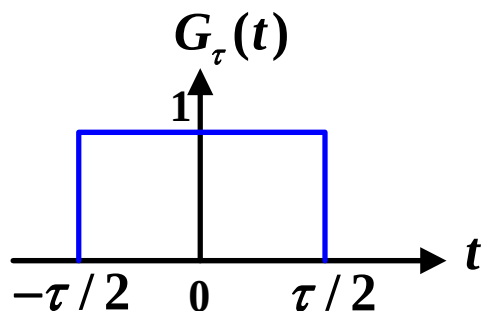
$$1 \Leftrightarrow 2\pi\delta(\omega)$$



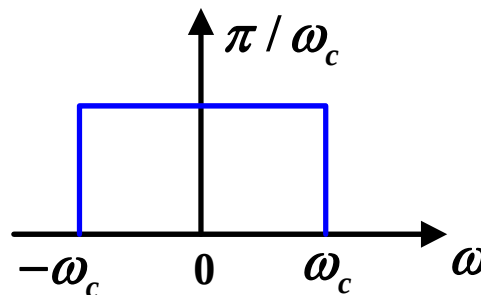
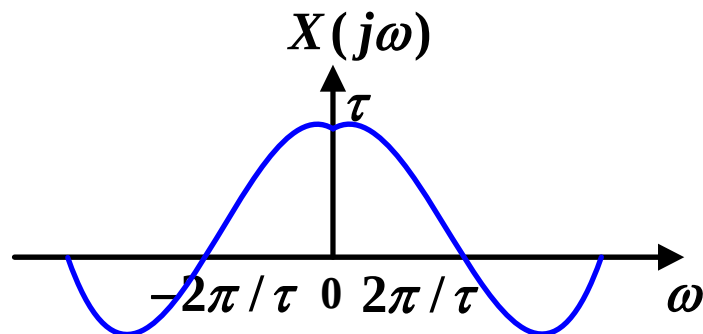
➤ FT与IFT的对偶性

II: $F(t) \leftrightarrow 2\pi f(-\omega)$

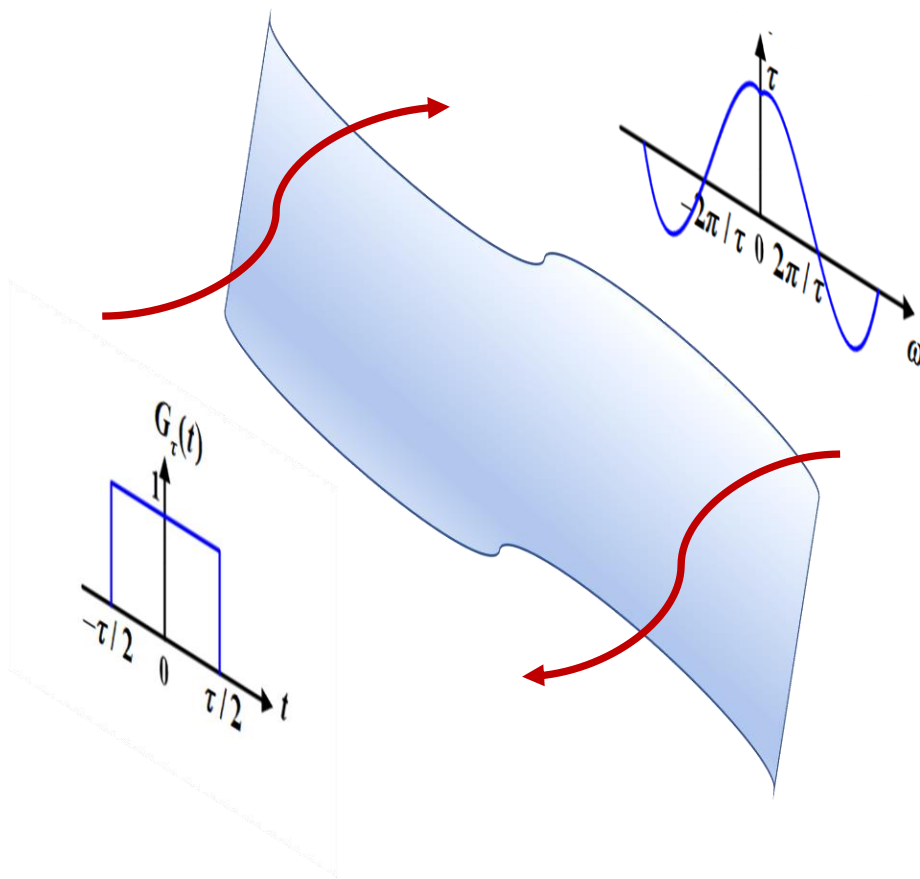
$$G_{\tau}(t) \leftrightarrow \tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$



$$\text{Sa}(\omega_c t) \leftrightarrow \frac{\pi}{\omega_c} G_{2\omega_c}(\omega)$$

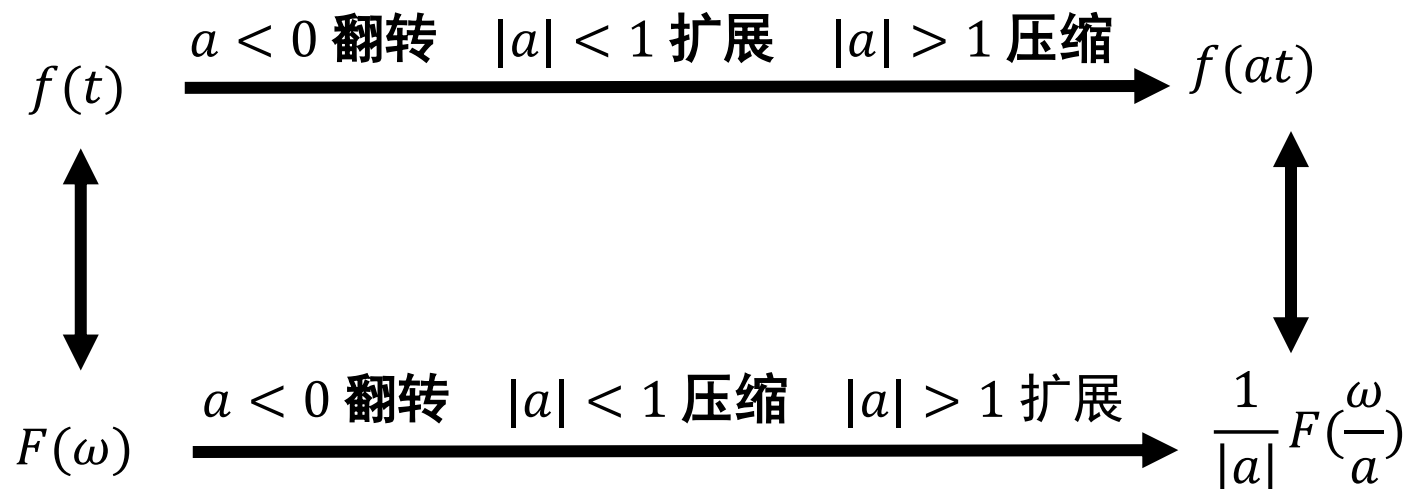


➤ FT与IFT的对偶性



➤ FT的尺度变换特性

$$F[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right), \quad (a \neq 0)$$



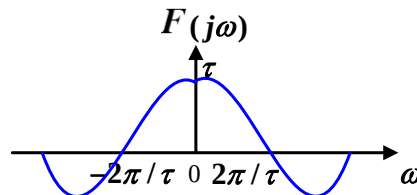
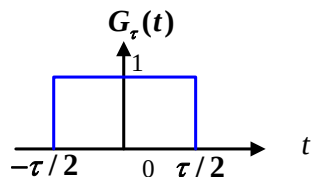
特例: $f(-t) \leftrightarrow F(-\omega)$

➤ FT的尺度变换特性

$$F[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right), \quad (a \neq 0)$$

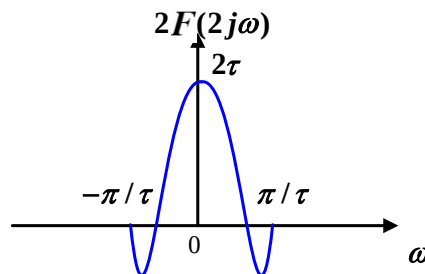
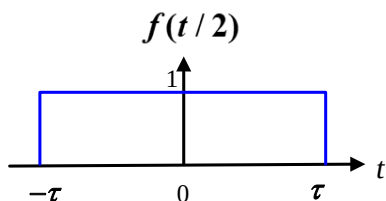
测不准

$$f(t) = G_\tau(t)$$



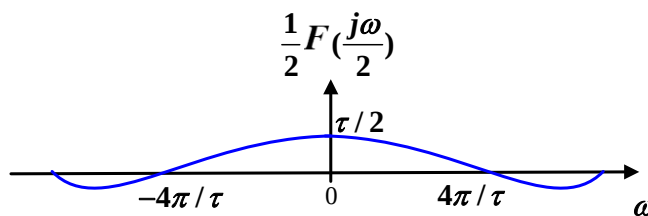
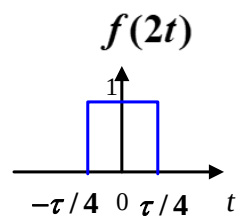
$$G_\tau(t) \leftrightarrow \tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

$$f(t/2)$$



$$G_{2\tau}(t) \leftrightarrow 2\tau \text{Sa}(\omega\tau)$$

$$f(2t)$$



若 $f(t)$ 为奇函数, 即 $f(-t) = -f(t)$,
则 $F(-\omega) = -F(\omega)$

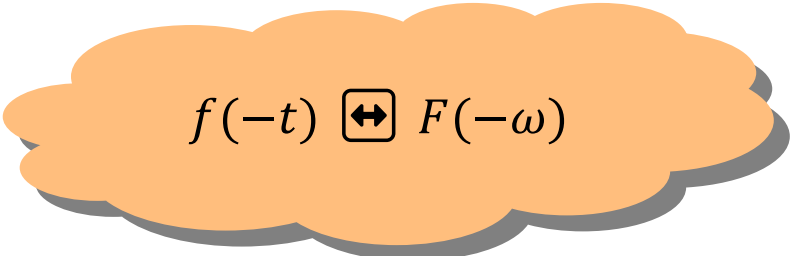
➤ FT的尺度变换特性

例 试求下列信号的频谱

$$e^{at}u(-t), \quad a > 0$$

$$e^{-at}u(t) + e^{at}u(-t) = e^{-a|t|}$$

$$e^{-at}u(t) - e^{at}u(-t)$$


$$f(-t) \leftrightarrow F(-\omega)$$

解：

$$e^{-at}u(t) \leftrightarrow \frac{1}{j\omega + a}, \quad a > 0$$

$$e^{at}u(-t) \leftrightarrow \frac{1}{a - j\omega}, \quad a > 0$$

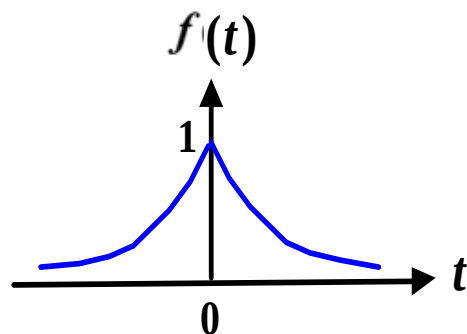
➤ FT的尺度变换特性

例 试求下列信号的频谱

$$e^{at}u(-t), \quad a > 0$$

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$$e^{-at}u(t) - e^{at}u(-t)$$



解: $e^{-at}u(t) \boxed{\leftrightarrow} \frac{1}{j\omega + a}, \quad a > 0$

$$e^{-at}u(t) + e^{at}u(-t) = e^{-a|t|} \quad \boxed{\leftrightarrow} \quad \frac{1}{a + j\omega} + \frac{1}{a - j\omega} = \underline{\underline{\frac{2a}{a^2 + \omega^2}}}$$

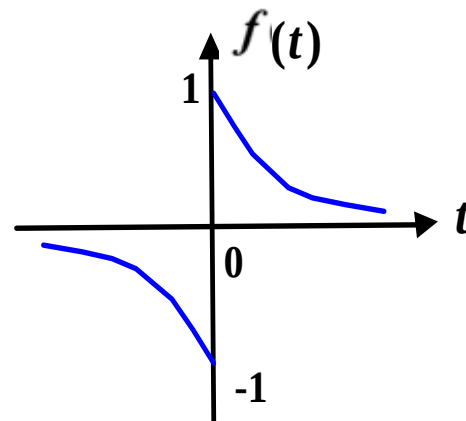
➤ FT的尺度变换特性

例 试求下列信号的频谱

$$e^{at}u(-t), \quad a > 0$$

$$e^{-at}u(t) + e^{at}u(-t) = e^{-a|t|}, \quad a > 0$$

$$e^{-at}u(t) - e^{at}u(-t), \quad a > 0$$



解: $e^{-at}u(t) \boxed{\leftrightarrow} \frac{1}{j\omega + a}, \quad a > 0$

$$e^{-at}u(t) - e^{at}u(-t) \boxed{\leftrightarrow} \frac{1}{a + j\omega} - \frac{1}{a - j\omega} = \frac{-2j\omega}{a^2 + \omega^2}$$

➤ FT的时移特性

若 $f(t) \leftrightarrow F(\omega)$, 则 $f(t + t_0) \leftrightarrow F(\omega)e^{j\omega t_0}$ t_0 为任意实数

证明: $f(t + t_0) \leftrightarrow \int_{-\infty}^{\infty} f(t + t_0)e^{-j\omega t} dt$

令 $t + t_0 = \tau$

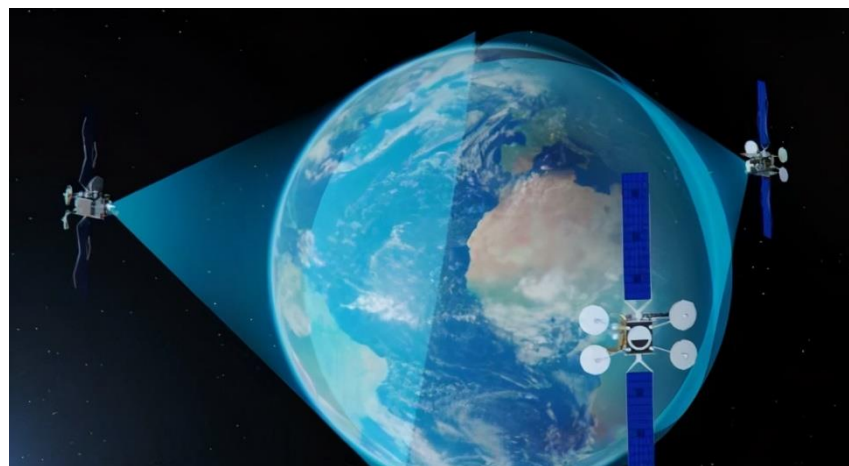
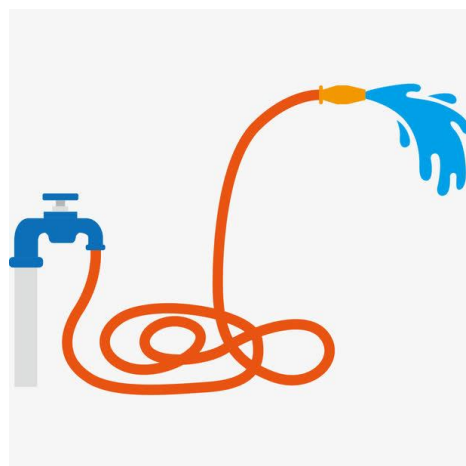
$$\begin{aligned} &= \int_{-\infty}^{\infty} f(\tau)e^{-j\omega(\tau-t_0)} d\tau \\ &= e^{j\omega t_0} \int_{-\infty}^{\infty} f(\tau)e^{-j\omega\tau} d\tau \\ &= e^{j\omega t_0} F(\omega) \end{aligned}$$

➤ FT的时移特性

若 $f(t) \leftrightarrow F(\omega)$, 则 $f(t+t_0) \leftrightarrow F(\omega)e^{j\omega t_0}$, t_0 为任意实数

意义: $f(t+t_0) \leftrightarrow |F(\omega)| e^{j[\angle F(\omega) + \omega t_0]}$

时域延时, 频域则是相位变化, 不影响幅度谱, 只在相位谱上叠加一个线性相位。



➤ FT的频移特性

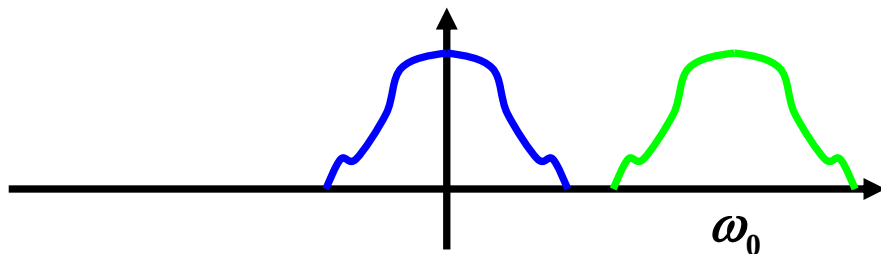
若 $f(t) \leftrightarrow F(\omega)$, 则

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

ω_0 为任意实数

证明:

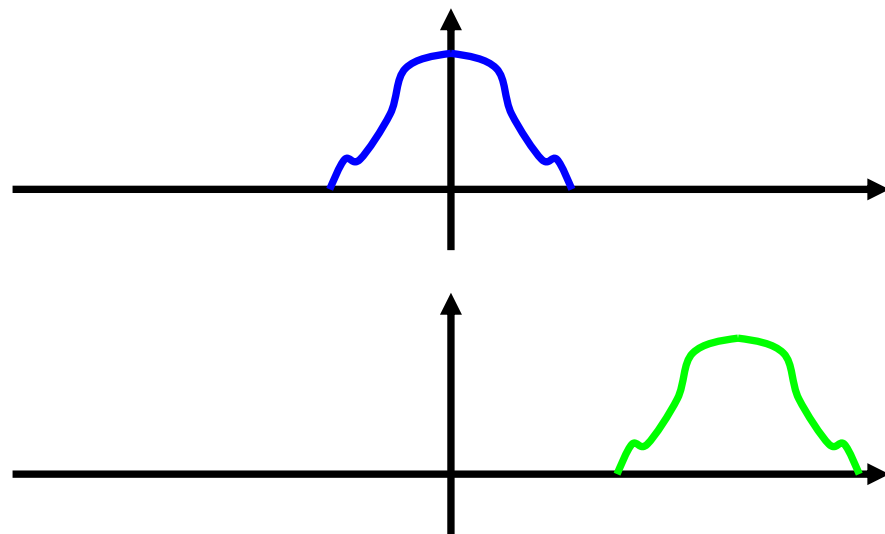
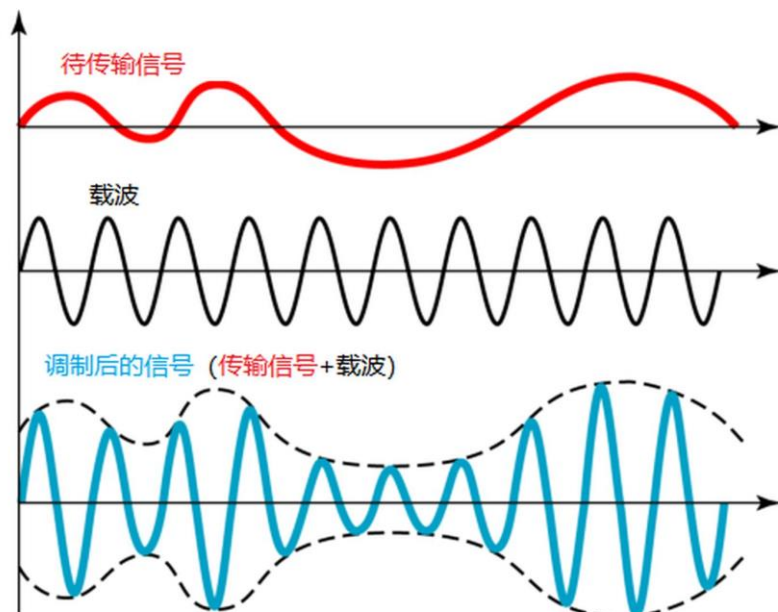
$$\begin{aligned} f(t)e^{j\omega_0 t} &\leftrightarrow \int_{-\infty}^{\infty} [f(t)e^{j\omega_0 t}] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) e^{-j(\omega - \omega_0)t} dt \\ &= F(\omega - \omega_0) \end{aligned}$$



➤ FT的频移特性

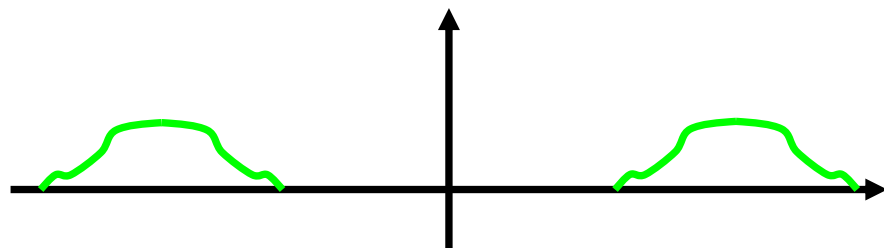
$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

信号的频率调制：调频



$$\mathcal{F}[f(t)\cos\omega_0 t] = \frac{1}{2}F(\omega + \omega_0) + \frac{1}{2}F(\omega - \omega_0)$$

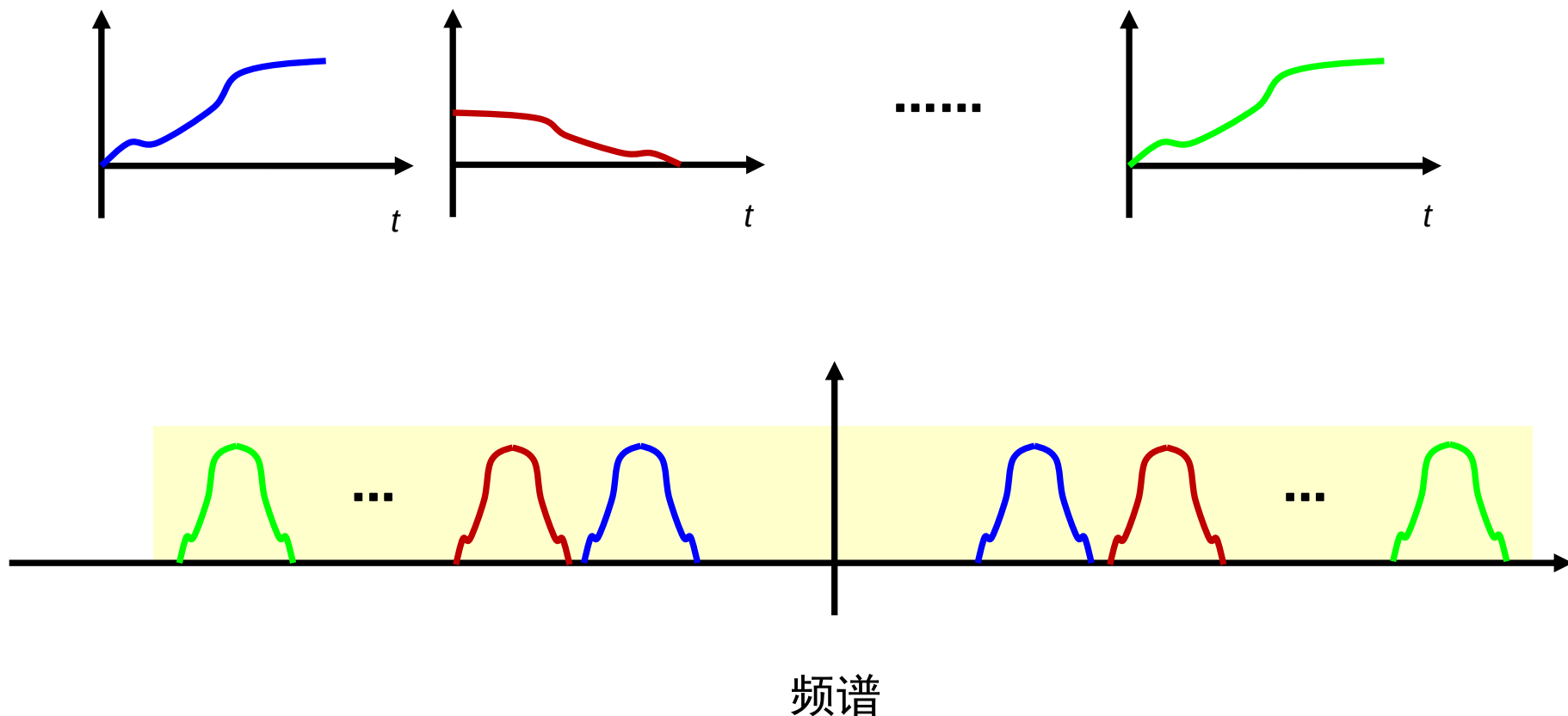
$$\cos\omega_0 t = (e^{j\omega_0 t} + e^{-j\omega_0 t}) / 2$$



➤ FT的频移特性

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$$

信号的频率调制：频分复用



➤ FT的微积分

- 微分特性:

$$\left\{ \begin{array}{ll} \text{时域微分} & \frac{d}{dt} f(t) \Leftrightarrow j\omega F(\omega) \\ \text{频域微分} & \frac{dF(\omega)}{d\omega} \Leftrightarrow -jtf(t) \end{array} \right.$$

- 积分特性:

$$\left\{ \begin{array}{ll} \text{时域积分} & \int_{-\infty}^t f(\tau) d\tau \Leftrightarrow (j\omega)^{-1} F(\omega) + \pi F(0)\delta(\omega) \\ \text{频域积分} & \int_{-\infty}^{\omega} F(\lambda) d\lambda \Leftrightarrow \pi f(0)\delta(t) + \frac{1}{-jt} f(t) \end{array} \right.$$

➤ FT的微积分

- 微分特性： 时域微分 $\frac{d}{dt} f(t) \Leftrightarrow j\omega F(\omega)$

证明： $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$

$$\frac{d}{dt} f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) j\omega e^{j\omega t} d\omega$$

$$\frac{d}{dt} f(t) \boxed{\Leftrightarrow} j\omega F(\omega)$$

推论： 若 $\frac{d^n f(t)}{dt^n}$ 存在, 则 $\frac{d^n f(t)}{dt^n} \boxed{\Leftrightarrow} (j\omega)^n F(\omega)$

➤ FT的微积分

- 微分特性： 频域微分 $\frac{dF(\omega)}{d\omega} \Leftrightarrow -jtf(t)$

证明：
$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

$$F'(\omega) = \int_{-\infty}^{\infty} f(t)(-jt)e^{-j\omega t} dt$$

$$-jtf(t) \boxed{\Leftrightarrow} \frac{d}{d\omega} F(\omega)$$

➤ FT的微积分

- 微分特性： 频域微分 $\frac{dF(\omega)}{d\omega} \Leftrightarrow -jtf(t)$

若 $f(t) \boxed{\leftrightarrow} F(\omega)$, 则 $tf(t) \boxed{\leftrightarrow} j \frac{d}{d\omega} F(\omega)$

$$t \boxed{\leftrightarrow} 2\pi j\delta'(\omega)$$

$$tu(t) \boxed{\leftrightarrow} j\pi\delta'(\omega) - \frac{1}{\omega^2}$$

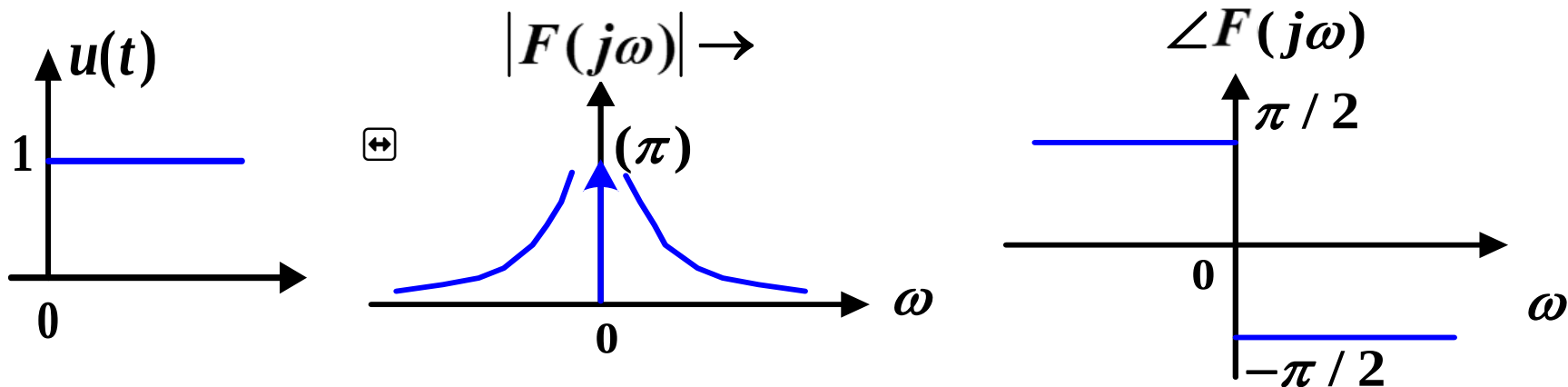
$$1 \boxed{\leftrightarrow} 2\pi\delta(\omega)$$

$$u(t) \boxed{\leftrightarrow} \pi\delta(\omega) + \frac{1}{j\omega}$$

➤ FT的微积分

- 积分特性: $\int_{-\infty}^t f(\tau) d\tau \Leftrightarrow (j\omega)^{-1} F(\omega) + \pi F(0) \delta(\omega)$

$$u(t) \boxed{\leftrightarrow} \pi \delta(\omega) + \frac{1}{j\omega}$$



➤ FT的卷积定理

- 时域卷积定理:

$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$$

- 频域卷积定理:

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} \mathcal{F}[f_1(t)] * \mathcal{F}[f_2(t)]$$

➤ FT的卷积定理

• 时域卷积定理：

$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$$

$$\mathcal{F}[f_1(t) * f_2(t)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau \right] e^{-j\omega t} dt \quad (\text{卷积和FT的定义})$$

$$= \int_{-\infty}^{\infty} f_1(\tau) \left[\int_{-\infty}^{\infty} f_2(t - \tau) e^{-j\omega t} dt \right] d\tau \quad (\text{交换积分次序})$$

$$= \int_{-\infty}^{\infty} f_1(\tau) \left[\mathcal{F}[f_2(t)] e^{-j\omega\tau} \right] d\tau \quad (\text{FT定义及其时移特性})$$

$$= \mathcal{F}[f_2(t)] \int_{-\infty}^{\infty} f_1(\tau) e^{-j\omega\tau} d\tau \quad (\text{常函数提出来})$$

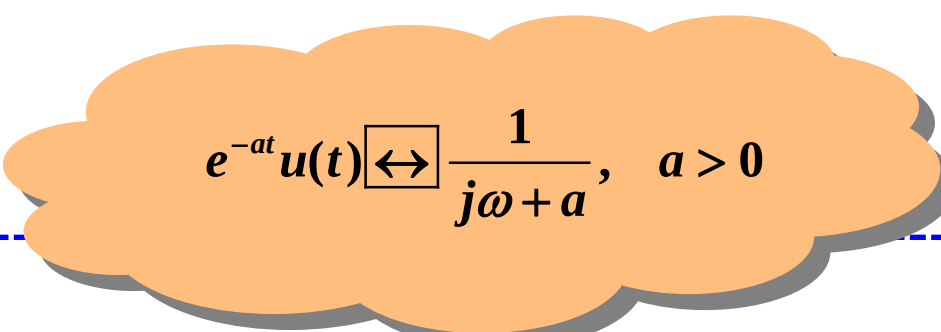
$$= \mathcal{F}[f_2(t)] \cdot \mathcal{F}[f_1(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)] \quad (\text{FT定义})$$

➤ FT的卷积定理

- 时域卷积定理:

$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$$

例 试求 $[e^{-t}u(t)] * [e^{-2t}u(t)]$


$$e^{-at}u(t) \boxed{\leftrightarrow} \frac{1}{j\omega + a}, \quad a > 0$$

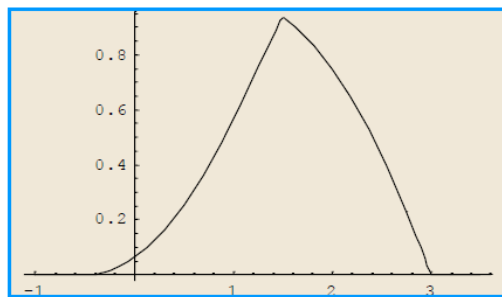
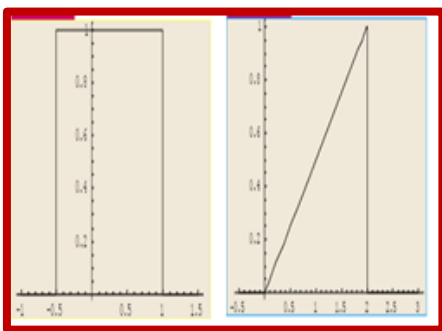
解:

$$\begin{aligned} [e^{-t}u(t)] * [e^{-2t}u(t)] &\boxed{\leftrightarrow} \frac{1}{1+j\omega} \cdot \frac{1}{2+j\omega} \\ &= \frac{1}{1+j\omega} + \frac{-1}{2+j\omega} \end{aligned}$$

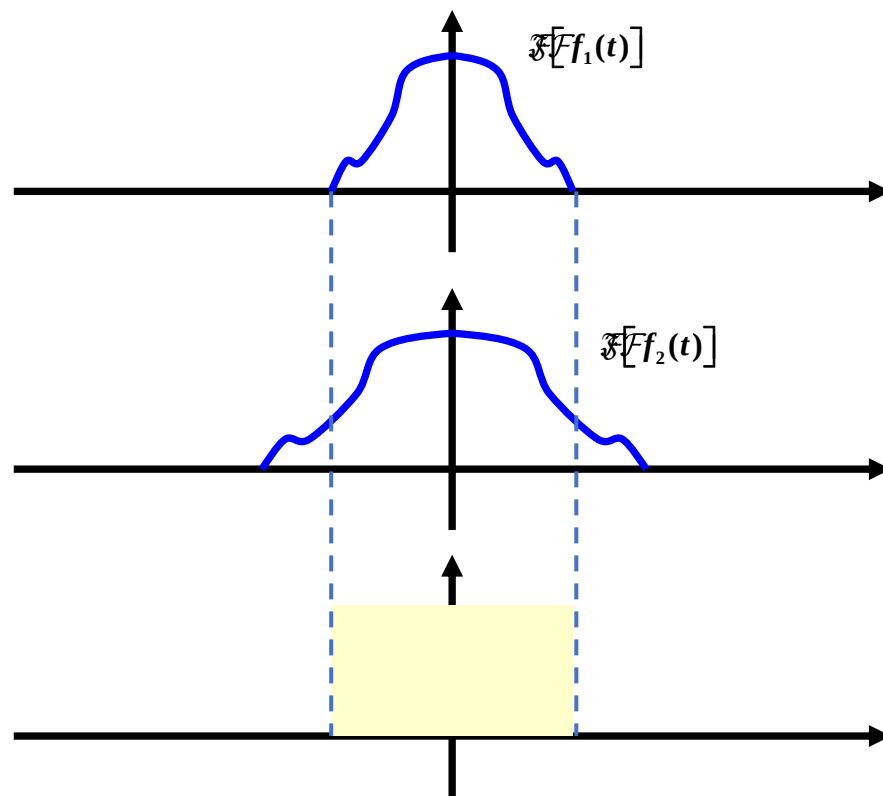
$$[e^{-t}u(t)] * [e^{-2t}u(t)] = e^{-t}u(t) - e^{-2t}u(t)$$

➤ FT的卷积定理

- **时域卷积定理：** $\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$



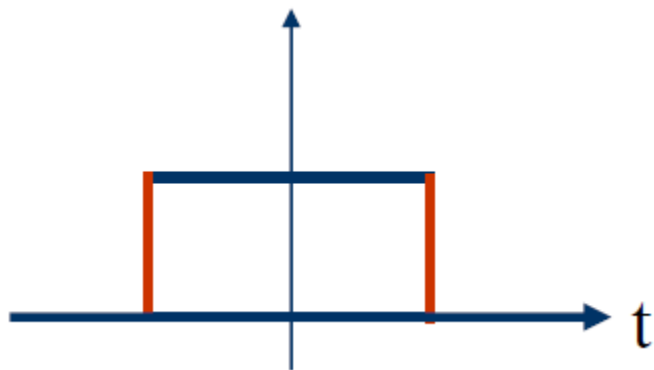
时域：长度增加



频域：长度不增加（不产生新的频率成分）

➤ FT的卷积定理

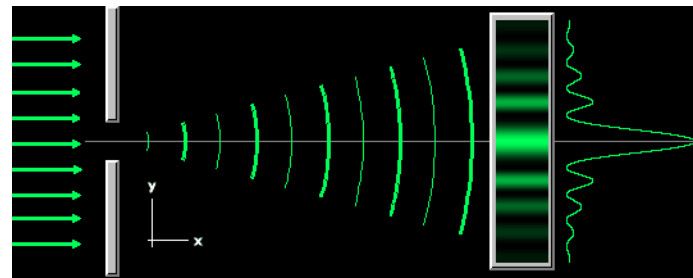
- **频域卷积定理：** $\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} \mathcal{F}[f_1(t)] * \mathcal{F}[f_2(t)]$
- 信号截取时，是使用如下矩形窗相乘来实现的：



$$G_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & \text{其它} \end{cases}$$
$$EG_{\tau}(t) \leftrightarrow E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

因此，矩形信号边缘的跳变将引起原信号的频谱会产生畸变：

$$F(\omega) = \frac{1}{2\pi} F_1(\omega) * F_2(\omega) = \frac{1}{2\pi} F_1(\omega) * \tau \text{Sa}\left(\frac{\tau}{2} \omega\right)$$



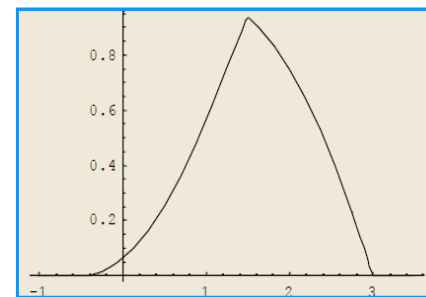
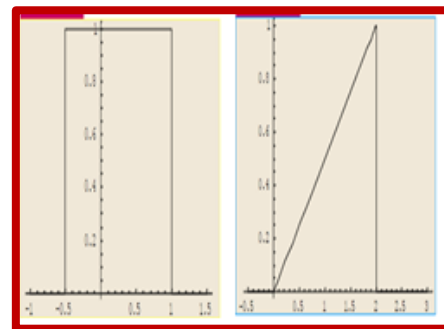
➤ FT的卷积定理总结

- 时域卷积定理:

$$\mathcal{F}[f_1(t) * f_2(t)] = \mathcal{F}[f_1(t)] \cdot \mathcal{F}[f_2(t)]$$

- 频域卷积定理:

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} \mathcal{F}[f_1(t)] * \mathcal{F}[f_2(t)]$$



两个信号卷积，卷不出新的频率成分出来。反之，两个信号相乘，可以得到新的频率成分。

➤ 信号的能量与频谱的关系（帕斯瓦尔定理）

$$\int_{-\infty}^{\infty} \|f(t)\|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|F(\omega)\|^2 d\omega$$

$$\begin{aligned} \int_{-\infty}^{\infty} \|f(t)\|^2 dt &= \int_{-\infty}^{\infty} f(t) f^*(t) dt \\ &= \int_{-\infty}^{\infty} f(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \right]^* dt && \text{(IFT定义)} \\ &= \int_{-\infty}^{\infty} f(t) \cdot \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} F^*(\omega) e^{-j\omega t} d\omega \right) dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F^*(\omega) \cdot \left(\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \right) d\omega && \text{(交换积分次序)} \\ &= \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} F^*(\omega) F(\omega) d\omega \right) && \text{(FT定义)} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \|F(\omega)\|^2 d\omega \end{aligned}$$

即

$$\int_{-\infty}^{\infty} \|f(t)\|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \|F(\omega)\|^2 d\omega$$

➤ FT性质总结

1. 揭示时域与频域的对应关系，进一步理解FT意义
2. 便于求解FT，在记住基本傅氏变换对基础上，灵活运用性质
3. 卷积定理地位突出

$$f_1(t) * f_2(t) \leftrightarrow F_1(\omega) \cdot F_2(\omega)$$

$$f_1(t) \cdot f_2(t) \leftrightarrow \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

➤ FT性质总结

① 时移 $f(t+t_0) = f(t) * \delta(t+t_0) \quad \boxed{\leftrightarrow} \quad F(\omega)e^{j\omega t_0}$

② 频移 $f(t)e^{j\omega_0 t} \quad \boxed{\leftrightarrow} \quad \frac{1}{2\pi} F(\omega) * [2\pi\delta(\omega-\omega_0)]$
 $= F(\omega-\omega_0)$

③ 时域微分 $f'(t) = f(t) * \delta'(t) \quad \boxed{\leftrightarrow} \quad F(\omega) \cdot j\omega$

④ 频域微分 $tf(t) \quad \boxed{\leftrightarrow} \quad \frac{1}{2\pi} F(\omega) * [j2\pi\delta'(\omega)] = jF'(\omega)$

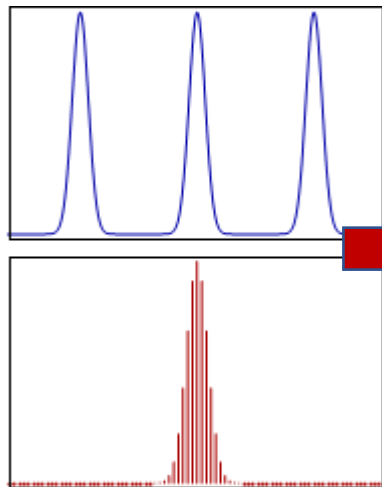
⑤ 时域积分 $\int_{-\infty}^t f(\tau)d\tau = f(t) * u(t) \quad \boxed{\leftrightarrow} \quad \pi F(0)\delta(\omega) + \frac{F(\omega)}{j\omega}$

傅里叶变换的性质

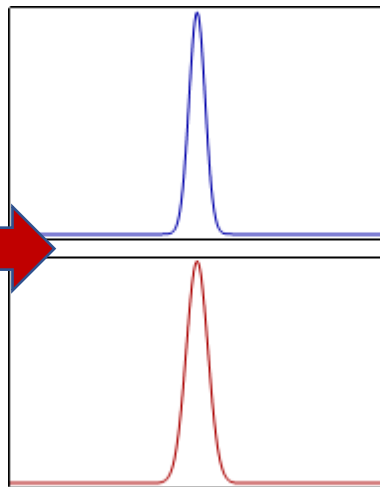
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周期

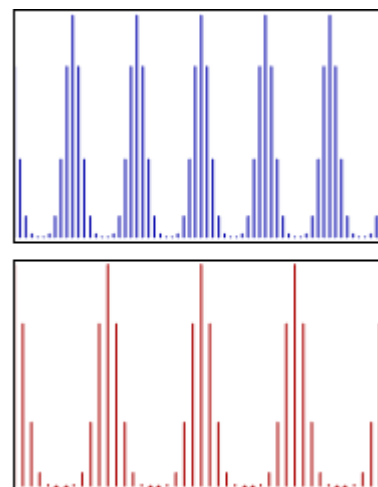
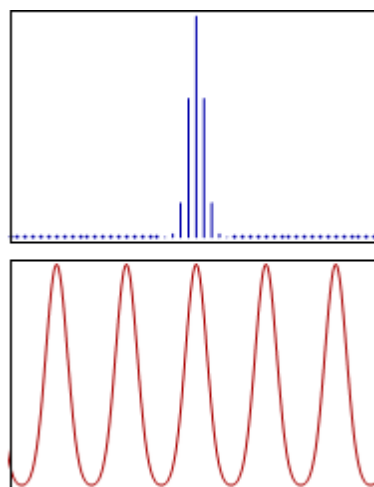
非周期



FS



FT



周期信号的傅里叶变换

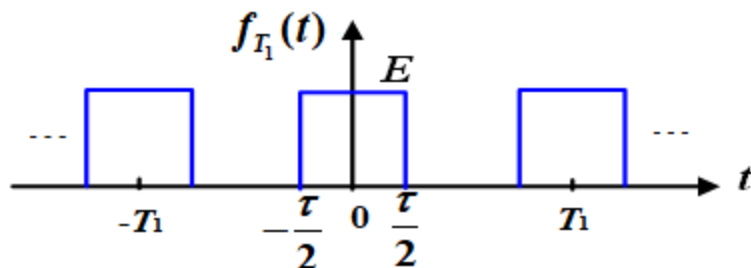
回顾：非周期信号

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

FT存在的充分条件：**绝对可积**

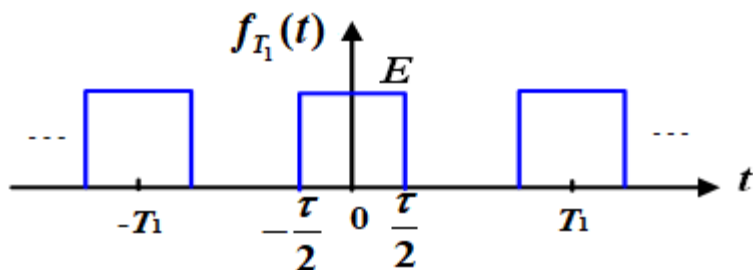
$$\int_{-\infty}^{+\infty} |f(t)| dt < \infty$$

周期信号

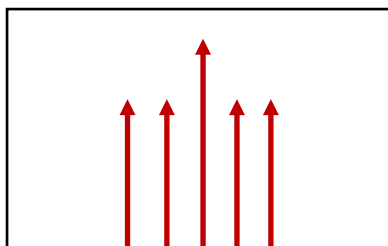
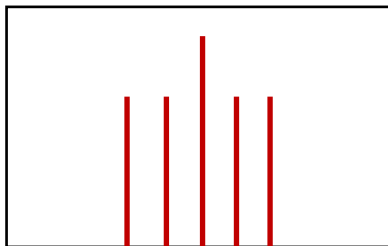


$$\int_{-\infty}^{+\infty} |f_{T_1}(t)| dt \rightarrow \infty$$

周期信号



$$\int_{-\infty}^{+\infty} |f_{T_1}(t)| dt \rightarrow \infty$$



周期信号的傅里叶变换

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例 求 $e^{j\omega_0 t}$ 、 $\cos \omega_0 t$ 及 $\sin \omega_0 t$ 的频谱

解：

$$1 \leftrightarrow 2\pi\delta(\omega)$$

$$e^{j\omega_0 t} \boxed{\leftrightarrow} 2\pi\delta(\omega - \omega_0)$$

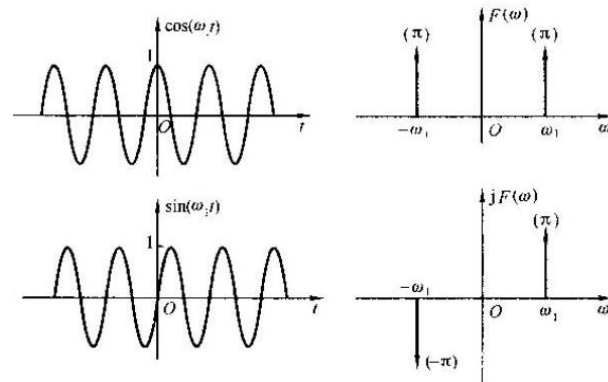
$$\cos \omega_0 t \stackrel{\text{Euler公式}}{=} \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$$

$$\cos \omega_0 t \boxed{\leftrightarrow} \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

$$\begin{aligned} \sin \omega_0 t &\stackrel{\text{Euler公式}}{=} \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}] \\ &= \frac{j}{2} [e^{-j\omega_0 t} - e^{j\omega_0 t}] \end{aligned}$$

$$\sin \omega_0 t \boxed{\leftrightarrow} j\pi\delta(\omega + \omega_0) - j\pi\delta(\omega - \omega_0)$$

FT的频移特性

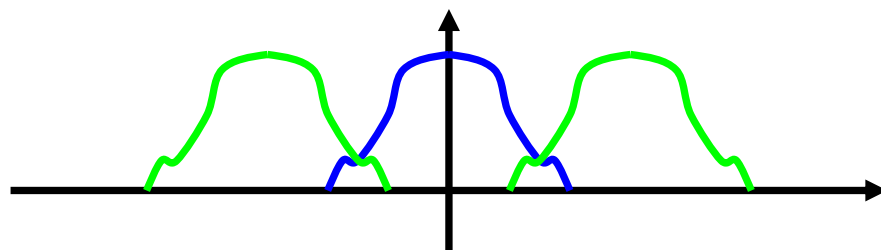
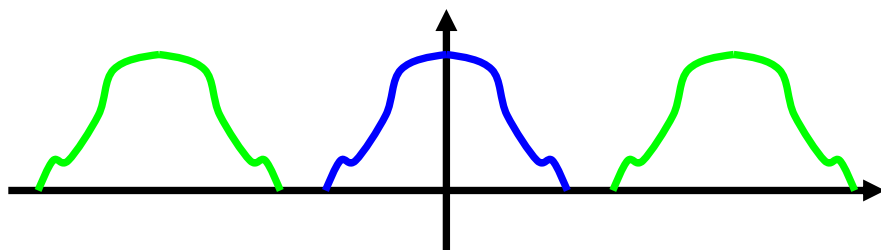


- 理论上，时域信号乘以一个复指数信号，原信号的频谱将被搬移到复指数信号的频率处。
- 实际应用中，利用欧拉公式，通过乘以正弦或余弦信号，可以达到**频谱搬移**的目的。

$$F[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0)$$

$$\cos \omega_0 t \leftrightarrow \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

$$\sin \omega_0 t \leftrightarrow j\pi\delta(\omega + \omega_0) - j\pi\delta(\omega - \omega_0)$$



$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t}$$

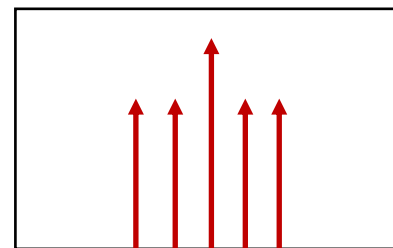
$$\mathcal{F}[f(t)] = \sum_{n=-\infty}^{\infty} F_n \mathcal{F}[e^{jn\omega_1 t}]$$

$$\mathcal{F}[e^{jn\omega_1 t}] = 2\pi\delta(\omega - n\omega_1)$$

$$\mathcal{F}[f(t)] = \sum_{n=-\infty}^{\infty} F_n 2\pi\delta(\omega - n\omega_1)$$

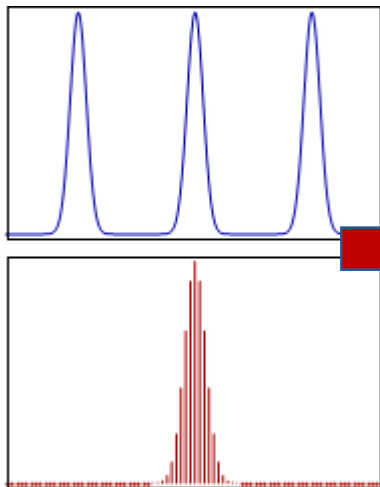
$$F(n\omega_1) = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) e^{-jn\omega_1 t} dt$$

- 周期信号频谱是离散谱。
- 但傅里叶变换的频谱是频谱密度，级数才是幅度值
- 因此每个谐频点的密度是无穷大
- 但频谱幅度则是有限的（无穷大密度乘上无穷小的频率范围即谐频点得到有限值的幅值）

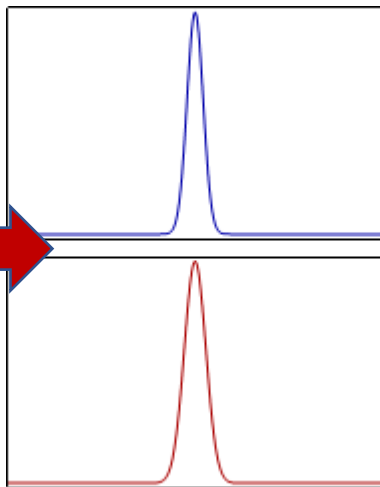


周期

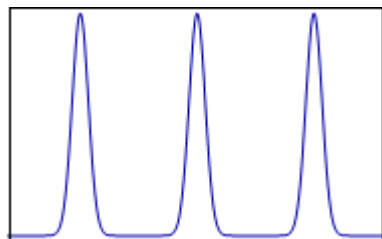
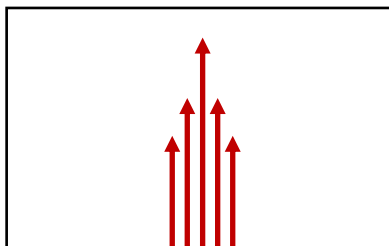
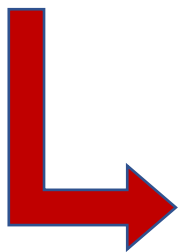
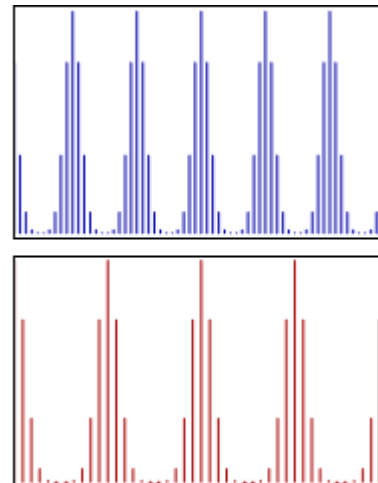
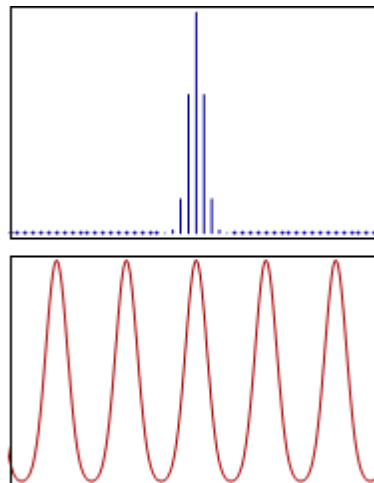
非周期



FS



FT

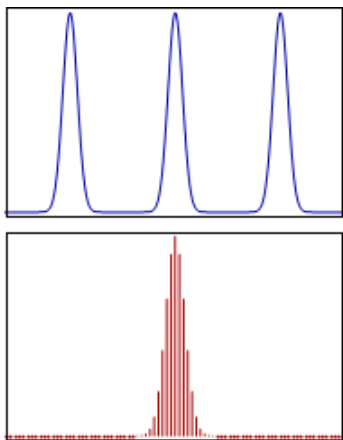


$$f(t) = \sum_{i=1}^{\infty} c_i \varphi_i(t)$$

Why?

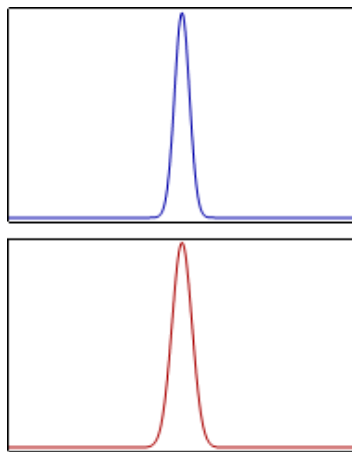


连续周期信号



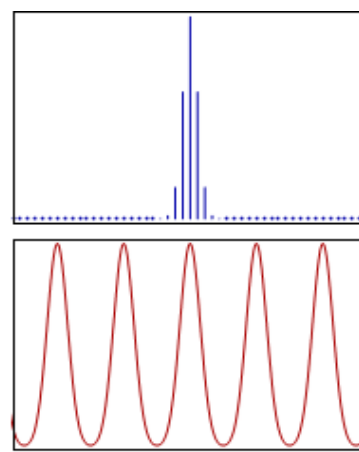
$$\{e^{jn\omega_1 t}\}$$

连续信号

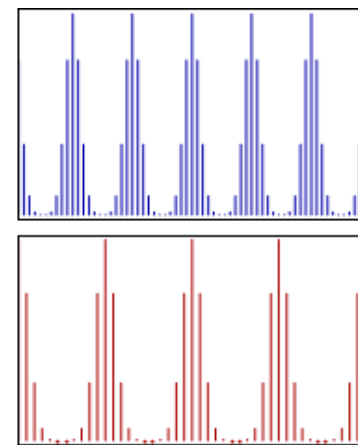


$$\{e^{j\omega t}\}$$

离散信号



离散周期信号



作业1：写出指数函数 ($f(t) = e^{-at}$, $a > 0, t > 0$) 的傅里叶变换。

作业2：

已知 $f(t) = \begin{cases} t & 0 \leq t < \tau \\ \tau & \tau \leq t < 2\tau \\ 0 & t < 0 \text{ 或 } t \geq 2\tau \end{cases}$, 求该函数的傅里叶变换。